

Exact Reductions for the Pythagorean–Burrau Nonperiodicity Conjecture

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Abstract

We study free-fall solutions of the planar Newtonian three-body problem in which the masses and their opposite side lengths are the same Pythagorean triple. The desired nonperiodicity theorem is not proved in this draft. We give exact reductions to primitive triples and a rational algebraic parameter curve, prove equivalence with absence of a second brake, construct a syzygy-valid three-component reduced brake residual, derive the universal zero-angular-momentum reduced system, and verify exact initial identities. The Lagrange–Jacobi identity further reduces every possible second brake to an explicitly collision-separated strict maximum of the moment of inertia and gives a finite validated-cover criterion. This criterion has certified five named primitive triples, but it does not yet control all maximum-event branches over the parameter curve. We also prove that the real isosceles-right endpoint has no second brake and ends in finite collision. At the opposite singular endpoint, a Levi–Civita argument proves that every sufficiently skinny member avoids the limiting heavy-pair collision on its first encounter, with explicit miss-distance and angular-momentum asymptotics. We derive the later rectilinear restricted scattering limit and prove that it has a parabolic separatrix, ruling out a phase-uniform strict-escape shortcut. We derive the exact collapsed-pair phase clock and the finite-parameter field through second order, exposing a parabolic time-translation gauge in the candidate splitting integral. A rigorous interval-Taylor computation proves the remaining finite Jacobi sign uniformly on an explicit launch-speed interval. An analytic compactification then verifies McGehee’s degenerate stable-manifold hypotheses and proves that the limiting incoming and outgoing parabolic curves cross transversely. Radius-dependent tail estimates and exact Levi–Civita normal invariants then match the true finite-parameter state to the restricted incoming curve. Consequently infinitely many open skinny parameter intervals, and hence infinitely many primitive Pythagorean triples, are nonperiodic. An exact equidistribution argument strengthens this to a positive-lower-density subset of the explicit family $(4n^2 - 1, 4n, 4n^2 + 1)$. On the complementary restricted side we prove infinitely many first-turn brake-state resonances, showing that finite-parameter resonance splitting, rather than a uniform limiting no-brake lemma, is required. We then validate the transverse scattering sign and prove that normalized outer angular momentum stays nonzero at the first outer turn uniformly throughout this accumulating resonance layer. For the next compact encounter we prove a positive centered-parabolic endpoint, and at its opposite triple-collision endpoint we prove uniqueness and collision safety of the universal peel-off base and validate its positive slow-mode first-turn transfer. In the remaining planar joint-limit family, an analytic stable-tail enclosure and correlated interval-Newton computations prove two distinct transverse light–heavy collision parameters, one for each primary, in explicit local boxes. An exact collision-adapted finite-mass reduction and a differentiated two-mode inclination theorem continue this root to a normalized collision graph through the logarithmic saddle dwell; the exact tied curve is reduced locally to one scalar signed gap from that graph. Combining its continuity with the exact rapidly winding skinny clock proves that infinitely many distinct real tied parameters have a finite-time classical collision, though none of these isolated parameters is proved rational. The certified limiting collision–ejection continuation also crosses the following

heavy-binary collision into a strict escape cone. A hierarchical Jacobi-energy bootstrap transfers this to a finite-mass local collision-or-escape theorem. Hence every sufficiently late phase bracket either contains an earlier collision or contains an open tied collision-or-escape interval. The full conjecture remains open because the compact phase interior, the one-parameter planar scattering family exposed by the finite-mass triple-endpoint blow-up, and later turns are not yet controlled. A bounded catalog audit finds a close numerical periodic brake orbit on the 3:4:5 mass slice, but its initial triangle is not the exact tied right triangle.

1 Exact conjecture

Let $a^2 + b^2 = c^2$ with positive integers, set $(m_1, m_2, m_3) = (a, b, c)$, and put

$$q_1 = (-c/2, 0), \quad q_2 = (c/2, 0), \quad q_3 = ((b^2 - a^2)/(2c), ab/c), \quad \dot{q}_i = 0.$$

The side opposite each mass has the same length as that mass. The conjecture asserts that no classical solution has a collision-free labelled period. Returns modulo rotation or label exchange and regularized continuation after collision are excluded.

2 Scaling and Euclid parameter

Lemma 1 (Simultaneous mass-length scaling). *If $q_i(t)$ solves the equations with masses m_i , then for $k > 0$*

$$\tilde{m}_i = km_i, \quad \tilde{q}_i(t) = kq_i(t/k)$$

solves the rescaled problem. A labelled period T corresponds exactly to a period kT .

Proof. Both sides of Newton's acceleration equation acquire the factor k^{-1} . Velocity is unchanged after the corresponding time substitution, and mutual distances acquire the positive factor k . $\square$

It follows that only primitive triples matter. For $u \in (0, 1)$ define

$$A(u) = \frac{1 - u^2}{1 + u^2}, \quad B(u) = \frac{2u}{1 + u^2}.$$

If $u = p/q$ is reduced, Euclid's integer triple is $(q^2 - p^2, 2pq, q^2 + p^2)$ and its common divisor is two exactly when p, q are both odd. Conversely Euclid's theorem recovers every ordered primitive triple. The inverse is $u = B/(1 + A)$, and leg exchange is the involution $u \mapsto (1 - u)/(1 + u)$. Its fixed point is $\sqrt{2} - 1$, giving the fundamental interval $0 < u \leq \sqrt{2} - 1$ for the real family.

After normalizing $c = 1$, masses and positions are

$$(A, B, 1), \quad q_1 = (-1/2, 0), \quad q_2 = (1/2, 0), \quad q_3 = ((B^2 - A^2)/2, AB).$$

3 Second brakes and a quotient residual

Lemma 2 (Second-brake lemma). *A classical free-fall segment generates a collision-free labelled periodic solution if and only if all inertial velocities vanish at some positive time before collision.*

Proof. Uniqueness and time reversal give $q(-t) = q(t)$. A brake at τ gives $q(\tau + s) = q(\tau - s)$ and hence a labelled period 2τ . Conversely, a labelled period T combined with the symmetry at zero gives reflection symmetry about $T/2$, and therefore zero velocity there. The period 2τ need not be minimal unless τ is the first positive brake. $\square$

In the center-of-mass frame define Jacobi vectors

$$X = q_2 - q_1, \quad Y = q_3 - \frac{m_1 q_1 + m_2 q_2}{m_1 + m_2}$$

and Hopf invariants

$$h(X, Y) = \left(\frac{|X|^2 - |Y|^2}{2}, X \cdot Y, X \times Y \right).$$

We use the quotient residual

$$\mathcal{B}(u, t) = \frac{d}{dt} h(X(u, t), Y(u, t)).$$

Away from triple collision, Dh has rank three and kernel equal to the common rotation direction. Since total linear and angular momentum vanish, $\mathcal{B} = 0$ is equivalent to vanishing of every labelled inertial velocity. Unlike three mutual-distance derivatives, this remains true at syzygy.

4 Maximum-event reduction

Let

$$I = \mu_1 |X|^2 + \mu_2 |Y|^2, \quad \sigma = \mu_1 \bar{X} \dot{X} + \mu_2 \bar{Y} \dot{Y}, \quad \zeta = \mu_1 \bar{X} \dot{X} - \mu_2 \bar{Y} \dot{Y}.$$

Here complex multiplication identifies the oriented plane with $\mathbb{C}$. Then $\dot{I} = 2 \operatorname{Re} \sigma$ and the angular momentum is $L = \operatorname{Im} \sigma$. Consequently, at $L = 0$ and away from $Y = 0$,

$$\dot{I} = 0 \text{ and } \zeta = 0 \iff \dot{X} = \dot{Y} = 0.$$

The implication from a brake to these three scalar equalities is unconditional. At the Jacobi degeneracy $Y = 0$, the Hopf residual $\mathcal{B}$ above supplies the coordinate-independent converse.

Theorem 3 (Brake-event identities). *Along a collision-free free-fall solution with initial potential U_0 , every brake satisfies*

$$K = 0, \quad U = U_0, \quad \ddot{I} = -2U_0 < 0, \quad r_{ij} \geq \frac{m_i m_j}{U_0}.$$

At any zero of $\dot{I}$ for which $\ddot{I} \geq 0$, one instead has $K \geq U_0$.

Proof. With positive potential magnitude U and energy $H = K - U = -U_0$, the Lagrange–Jacobi identity is

$$\ddot{I} = 4K - 2U = 2K - 2U_0 = 2U - 4U_0.$$

At a brake $K = 0$, giving the first three assertions. Since every summand $m_i m_j / r_{ij}$ is positive and bounded above by $U = U_0$, the separation bound follows. If $\ddot{I} \geq 0$, the middle expression instead gives $K \geq U_0$. $\square$

Corollary 4 (Validated-cover criterion). *Suppose a finite sequence of validated flow enclosures covers a classical trajectory segment and every enclosure proves at least one of*

$$\dot{I} \neq 0, \quad K > 0, \quad \mathcal{B} \neq 0.$$

Then that segment contains no brake. If its terminal enclosure satisfies a proved forward collision-or-escape criterion that permits no later brake, the initial free-fall solution is nonperiodic.

This criterion has been implemented with exact rational initial data and MPFR interval arithmetic. The archived certificates prove the conjecture for

$$(21, 20, 29), (5311, 5280, 7489), (8319, 8200, 11681), (33439, 32400, 46561), (72, 65, 97)$$

in both leg orderings. These are five point theorems, not a parameter-cover argument.

For each fixed u the separation bound is positive, and it is uniform on every compact parameter interval bounded away from $u = 0$. It degenerates in the skinny limit and is not a parameter-uniform Euclidean separation theorem on the whole open interval.

On any regular strict-maximum branch, the implicit-function theorem locally gives $t = t_k(u)$. The unresolved condition becomes the planar origin-avoidance problem

$$u \longmapsto \zeta(u, t_k(u)) \in \mathbb{C} \setminus \{0\}.$$

To turn this into a universal theorem one must also control branch creation, collision and escape endpoints, degenerate events, and the possible number of branches. The nominal dimension count alone supplies none of these facts.

5 Universal reduced system

Let

$$\mu_1 = \frac{m_1 m_2}{m_1 + m_2}, \quad \mu_2 = \frac{m_3(m_1 + m_2)}{m_1 + m_2 + m_3}.$$

Rotate so that $X = (R, 0)$ and write $Y = (P, Q)$. At zero angular momentum, Routh reduction gives

$$K_{red} = \frac{\mu_1 \dot{R}^2 + \mu_2 (\dot{P}^2 + \dot{Q}^2)}{2} - \frac{\mu_2^2 (P\dot{Q} - Q\dot{P})^2}{2[\mu_1 R^2 + \mu_2 (P^2 + Q^2)]}.$$

The positive potential is

$$U = \frac{m_1 m_2}{R} + \frac{m_1 m_3}{\sqrt{(P + \frac{m_2 R}{m_1 + m_2})^2 + Q^2}} + \frac{m_2 m_3}{\sqrt{(P - \frac{m_1 R}{m_1 + m_2})^2 + Q^2}}.$$

The Pythagorean curve on the brake boundary is

$$R = 1, \quad P = \frac{AB(B - A)}{A + B}, \quad Q = AB, \quad \dot{R} = \dot{P} = \dot{Q} = 0.$$

6 Exact initial geometry

The initial center of mass is the incenter because its barycentric weights are the opposite side lengths. With $I = \sum m_i |q_i - q_{CM}|^2$ and positive potential magnitude U , direct calculation gives

$$I_0 = abc, \quad U_0 = \frac{ab}{c} + \frac{ac}{b} + \frac{bc}{a}.$$

Thus after normalization

$$I_0 = AB, \quad U_0 = AB + (AB)^{-1}, \quad H = -U_0.$$

The configurational measure at fixed masses is

$$\mu_0 = U_0 \sqrt{I_0} = (AB)^{3/2} + (AB)^{-1/2},$$

and decreases strictly on the fundamental interval. It is not invariant under the simultaneous mass-length scaling used above.

For $D = r_{12}^2 - r_{23}^2 - r_{31}^2$, Newton's equations give

$$D(0) = \dot{D}(0) = 0, \quad \ddot{D}(0) = 2 \left(\frac{1}{AB} - AB(A+B) \right) > 0.$$

This is local only. Converged ordinary numerical experiments show D later changes sign for $u = 1/3$, so a global $D > 0$ cone is not pursued. An exact fourth-order calculation gives

$$D^{(4)}(0) = \frac{P_{20}(u)}{u^4(u-1)^4(u+1)^2(1+u^2)^4}.$$

Exact Sturm arithmetic shows that P_{20} has no zero on $(0, 83/200)$ and $P_{20}(0) = -1$. Hence $D^{(4)}(0) < 0$ throughout the fundamental interval. This uniform local sign still supplies no global cone.

7 The real isosceles-right endpoint

Theorem 5. *At $u = \sqrt{2} - 1$, the classical solution has no positive second brake and reaches a binary or triple collision in finite time.*

Proof. Here $m_1 = m_2 = m = 1/\sqrt{2}$ and reflection symmetry persists. Write the base bodies as $(-x, y_b), (x, y_b)$ and the apex as $(0, y_a)$, with $h = y_a - y_b$. Then

$$\ddot{x} = -\frac{m}{4x^2} - \frac{x}{(x^2 + h^2)^{3/2}} < 0.$$

Starting with $x_0 = 1/2$ and $\dot{x}(0) = 0$ gives $\dot{x} < 0$, excluding a second brake. While $x > 0$, one has $\ddot{x} \leq -m/(4x_0^2)$, whence

$$x(t) \leq x_0 - \frac{m}{8x_0^2} t^2.$$

Thus collision occurs no later than $\sqrt{8x_0^3/m} = 2^{1/4}$. $\square$

The endpoint is irrational, so this theorem does not settle an integer case. Its perturbation is singular because the collision unfolds into a close ordinary encounter when reflection symmetry is broken.

8 The first skinny encounter

Put $\epsilon = B$, $A = \sqrt{1 - \epsilon^2}$, and $M = A + 1$. For the tight-pair and outer Jacobi vectors

$$X = q_3 - q_1, \quad Y = q_2 - \frac{Aq_1 + q_3}{M},$$

rotate the encounter frame so that $X(0)/\epsilon = 1$ and scale

$$X = \epsilon x, \quad t = \frac{\epsilon^{3/2}}{\sqrt{M}} \tau.$$

The exact inner equation is

$$x_{\tau\tau} = -\frac{x}{|x|^3} + \frac{\epsilon^3}{M} \left[\Phi\left(y - \frac{A\epsilon}{M}x\right) - \Phi\left(y + \frac{\epsilon}{M}x\right) \right], \quad \Phi(v) = \frac{v}{|v|^3}.$$

The outer acceleration in τ -time is $O(\epsilon^3)$ on a fixed encounter interval. Introduce Levi-Civita variables

$$x = z^2, \quad d\tau = |z|^2 ds.$$

The augmented equations for z , z_s , the osculating Kepler energy, and the outer state are real analytic through $z = 0$. At $\epsilon = 0$,

$$z_0(s) = \cos(s/\sqrt{2}), \quad s_c = \pi/\sqrt{2}.$$

The first transverse parameter variation occurs at order five and satisfies

$$v_{ss} + \frac{1}{2}v = -\frac{3}{8}\cos^5(s/\sqrt{2}), \quad v(0) = v_s(0) = 0,$$

so that

$$v(s_c) = -\frac{15\pi}{128}.$$

Theorem 6 (First skinny miss). *There is $\epsilon_0 > 0$ such that for every real member with $0 < B = \epsilon < \epsilon_0$, the first tight-pair encounter is collision-free. At closest passage,*

$$r_{13,\min} = \frac{225\pi^2}{16384}\epsilon^{11}(1 + O(\epsilon)), \quad h_{13} = -\frac{15\pi}{64}\epsilon^{11/2} + O(\epsilon^{13/2}),$$

where $h_{13} = X \times \dot{X}$ is specific pair angular momentum.

Proof. Analytic dependence gives

$$\text{Im } z(s_c, \epsilon) = -\frac{15\pi}{128}\epsilon^5 + O(\epsilon^6).$$

Since $\partial_s \text{Re } z_0(s_c) = -1/\sqrt{2}$, choose a fixed neighborhood on which the real part remains strictly decreasing. Its unique zero has the displayed nonzero imaginary part. Outside that neighborhood z_0 is bounded away from zero up to a fixed outgoing section, while both outer separations are uniformly bounded below. Thus no collision occurs. Squaring the transverse displacement and converting the scaled angular momentum to physical time gives the two coefficients. $\square$

The first-encounter result alone does not prove skinny nonperiodicity. The binary remains bound with specific energy $-M/\epsilon + O(\epsilon^3)$, and the first-passage outer radial escape margin is $-2 + O(\epsilon)$. The outer body falls back on an order-one time scale, during which the binary executes $O(\epsilon^{-3/2})$ cycles. Later sections control this train and prove nonperiodicity on infinitely many phase windows, but not uniformly in phase.

Lemma 7 (Terminal escape certificate). *For a prospective binary of total mass M and an outer body of mass m_c , let r, ρ be the inner and outer Jacobi radii and $e = |\dot{x}|^2/2 - M/r$. Choose $\eta > 0$, set $R = M/\eta$ and $s_0 = \rho_0 - R$, and suppose*

$$s_0 > 0, \quad \dot{\rho}_0 > 0, \quad \delta = \frac{1}{2}\dot{\rho}_0^2 - \frac{M + m_c}{s_0} > 0,$$

$$e_0 + \frac{m_c \sqrt{2MR}}{\sqrt{2\delta} s_0^2} < -\eta.$$

Then the future solution either ends in inner binary collision or the outer body escapes with $\dot{\rho} \geq \sqrt{2\delta}$ forever. In particular there is no later classical brake.

Proof. On the bootstrap region $e < -\eta$, one has $r < R$ and $r|\dot{x}| < \sqrt{2MR}$. The Newton field derivative bound $\|D(v/|v|^3)\| \leq 2/|v|^3$ gives

$$|\dot{e}| \leq \frac{2m_c r |\dot{x}|}{(\rho - r)^3}.$$

Meanwhile

$$J = \frac{1}{2} \dot{\rho}^2 - \frac{M + m_c}{\rho - R}$$

is nondecreasing while $\dot{\rho} > 0$. Hence $\rho \geq \rho_0 + \sqrt{2\delta}(t - t_0)$, and the total future tidal increase of e is at most the allowance in the hypothesis. Its strict margin closes the bootstrap. $\square$

8.1 The late-plunge restricted limit

The terminal lemma cannot be applied uniformly over the later binary phase. Indeed, rescale near the outer plunge by

$$X = \epsilon R, \quad Y = \epsilon Z, \quad t = t_* + \epsilon^{3/2} \theta.$$

On compact nonsingular sets the exact Jacobi equations tend to

$$R_{\theta\theta} = -2 \frac{R}{|R|^3}, \quad Z_{\theta\theta} = -[\Phi(Z + R/2) + \Phi(Z - R/2)]. \quad (\text{LR})$$

After regularizing the prescribed binary, this convergence extends through its binary collisions provided the light particle is separate. The invariant rectilinear subspace has

$$R = (r, 0), \quad Z = (0, z), \quad \ddot{z} = -\frac{2z}{(z^2 + r^2/4)^{3/2}},$$

where r is the periodic collision–ejection binary with apocenter one. The outer instantaneous energy obeys

$$\frac{d}{d\theta} \left(\frac{1}{2} \dot{z}^2 - \frac{2}{\sqrt{z^2 + r^2/4}} \right) = \frac{r \dot{r}}{2(z^2 + r^2/4)^{3/2}}.$$

Theorem 8 (Restricted parabolic separatrix). *When the light particle crosses $z = 0$ at binary apocenter, there is a launch speed $v_* > 0$ for which it has no light–heavy collision and*

$$z(\theta) \rightarrow +\infty, \quad \dot{z}(\theta) \rightarrow 0$$

as $\theta \rightarrow +\infty$. Reflection and time reversal give a parabolic-to-parabolic scattering orbit.

Proof. For positive z , $\ddot{z} < 0$. Small launch speeds turn: with $r = \cos^2 \psi$ and $d\theta = \cos^2 \psi d\psi$, the zero-speed variational equation becomes, after $\zeta = \cos \psi w$,

$$w_{\psi\psi} + (1 + 14 \sec^2 \psi) w = 0.$$

Sturm comparison forces a simple zero before the first binary collision. Large launch speeds reach $z = 1$ while retaining a speed for which $\dot{z}^2/2 - 2/z > 0$; this quantity is thereafter nondecreasing. Let $\mathcal{T}$ be the nonempty open bounded set of speeds with a finite turn and put $v_* = \sup \mathcal{T}$. The boundary solution cannot turn and cannot have positive limiting speed, since either property is open. It cannot remain bounded, because the periodic attraction would then force a turn. $\square$

The binary collisions in (LR) are regularized features of an auxiliary limit, not physical continuations of a positive- ϵ orbit. Nevertheless the separatrix proves that a strict phase-uniform escape margin is impossible in the limiting problem. A finite- ϵ proof must instead match the incoming data and determine the separatrix splitting.

8.2 Reference phase and finite-parameter splitting

Put $A = \sqrt{1 - \epsilon^2}$, $M = 1 + A$, $N = M + \epsilon$, and let $\rho_0^2 = 2A^2/M$ be the exact initial outer Jacobi radius. Replacing the tight pair by a point mass gives the exact reference clock

$$\Phi_{\text{ref}} = \frac{\pi}{\epsilon^{3/2}} \sqrt{\frac{M}{N}} \rho_0^{3/2} = \pi \epsilon^{-3/2} - \frac{\pi}{4} \epsilon^{-1/2} - \frac{15\pi}{32} \epsilon^{1/2} + O(\epsilon^{3/2}). \quad (\text{RC})$$

Moreover $\Phi'_{\text{ref}} \sim -(3\pi/2)\epsilon^{-5/2}$. Thus every open phase arc has infinitely many reference preimage intervals accumulating at zero, each containing rational Euclid parameters. The matching theorem below transfers this reference statement to compact open windows of the true flow.

That long-cycle step can be proved without finite-eccentricity averaging. Fix $0 < \alpha < 1/6$ and stop at physical outer radius $\rho = \epsilon^\alpha$. Taylor cancellation in the outer Jacobi field gives a monopole remainder $O(\epsilon^2 \rho^{-4})$. Inner-energy and radial comparison estimates keep $|X| \leq 2\epsilon$ and give

$$t_\alpha - t_0(\epsilon^\alpha) = O(\epsilon^{2-3\alpha}).$$

In LC variables the scaled binary forcing is $O(\epsilon^{4-3\alpha})$ over regularized length $O(\epsilon^{-3/2})$. Harmonic-oscillator variation and the exact LC clock identity yield a regularized mean-anomaly error $O(\epsilon^{1-3\alpha})$. Hence, unless a prior classical collision has already ended the orbit, the renormalized incoming intercept satisfies

$$\text{dist}_{S^1}(\chi_\alpha, \Phi_{\text{ref}}) = O(\epsilon^{1/2-3\alpha}) = o(1). \quad (\text{PM})$$

Theorem 9 (Incoming-tail matching). *Fix a sufficiently large scaled incoming section Σ_K^- . As $\epsilon \rightarrow 0$, either a classical collision occurs before section reach or the exact regularized state $S_\epsilon(K)$ satisfies*

$$\text{dist}(S_\epsilon(K), \Gamma_K^-(\Phi_{\text{ref}}(\epsilon))) = o(1), \quad (\text{IT})$$

where $\Gamma_K^-(\chi)$ is the restricted incoming parabolic state with asymptotic mean-anomaly intercept χ .

Proof. Put $Y_e = \epsilon^{\alpha-1}$ with $0 < \alpha < 1/6$. The exact cancellation of the outer dipole leaves an $O(Y^{-4})$ remainder on the tail. If

$$\mathcal{H}_N = \frac{1}{2}|Z_\theta|^2 - \frac{N}{Y}, \quad \mathcal{L} = Z \times Z_\theta,$$

then the pre-plunge rates and a simultaneous energy-torque bootstrap give

$$|\mathcal{H}_N - E_{\text{ref},\epsilon}| \leq CY^{-3}, \quad |\mathcal{L}| \leq CY^{-3/2}, \quad -Y_\theta \asymp Y^{-1/2}. \quad (\text{TB})$$

These estimates are uniform for $K \leq Y \leq Y_e$.

Collision-uniform binary control uses exact LC first integrals rather than a singular Delaunay chart. With normalized energy E set

$$\mathcal{J} = (E + 1, L, K_{12}), \quad L = z_x p_y - z_y p_x, \quad K_{12} = p_x p_y - \frac{E}{2} z_x z_y.$$

On the radial LC circle,

$$\det \frac{\partial(L, K_{12})}{\partial(z_y, p_y)} = -\frac{1}{2}.$$

The three components are antipodal-invariant uniform normal coordinates, and direct differentiation of the forced LC equations gives $|\mathcal{J}_s| \leq C\epsilon Y^{-3}$. Complete LC blocks have uniformly comparable physical lengths; the two incomplete endpoint blocks are bounded directly in LC time. Consequently

$$|\mathcal{J}(Y)| \leq O(\epsilon^{5/2-3\alpha}) + C\epsilon Y^{-3/2},$$

and the accumulated binary-clock error is

$$O(\epsilon^{1-3\alpha/2}) + O(\epsilon |\log \epsilon|) = o(1). \quad (\text{BC})$$

Retain the exact finite- ϵ subtraction

$$F_\epsilon(Y) = n_{\text{bin}}(t_{\text{mono}} - t_0(\epsilon Y))$$

until Y is fixed. The reciprocal-speed consequence of (TB) and (BC) is

$$\text{dist}_{S^1}(\Lambda_\epsilon(Y) + F_\epsilon(Y), \chi_\epsilon) \leq CY^{-1/2} + o(1), \quad (\text{PG})$$

where $\chi_\epsilon = \Phi_{\text{ref}}(\epsilon) + o(1)$. The $CY^{-1/2}$ term retains the restricted quadrupole flight correction.

For any sequence $\epsilon_n \rightarrow 0$, LC compactness gives a diagonal limiting restricted tail. The inherited angular bounds are $Z_\perp = O(Y^{-1})$ and $Z'_\perp = O(Y^{-5/2})$. Reflection oddness factors the exact transverse equation as

$$\xi'' + (2Y^{-3} + O(Y^{-5}))\xi = 0.$$

With $x = \log Y$ this is an L^1 perturbation of a constant system whose modes are Y and $Y^{1/2}$; the inherited decay forces $\xi \equiv 0$. The limit is therefore the rectilinear incoming parabolic orbit. Taking $\epsilon_n \rightarrow 0$ at fixed Y in (PG), and only then $Y \rightarrow \infty$, identifies its intercept as the limit of $\Phi_{\text{ref}}(\epsilon_n)$. Uniqueness of the incoming McGehee graph proves (IT). $\square$

On the limiting symmetric orbit let $d^2 = z^2 + r^2/4$. Expansion of the exact finite-parameter Jacobi field gives

$$\begin{aligned} r'' &= -\frac{2}{r^2} - \epsilon \frac{r}{d^3} + \frac{\epsilon^2}{2r^2} + O(\epsilon^3), \\ Z''_y &= -\frac{2z}{d^3} - \epsilon \frac{z}{d^3} + \epsilon^2 \frac{z}{2d^3} + O(\epsilon^3), \quad Z''_x = \epsilon^2 \frac{3r^3}{16d^5} + O(\epsilon^3). \end{aligned} \quad (\text{FS})$$

The centered order- ϵ energy-transfer variation cancels by parity; longitudinal bending starts at order ϵ^2 . The first symmetry-allowed splitting direction is binary phase. Its naive whole-line Jacobi field is nevertheless gauge-degenerate: choosing the global time-translation field $\zeta = z'$ makes the proposed integrand exactly

$$\frac{d}{d\theta} \left(\frac{rr'}{2d^3} \right).$$

Because $z' = O(|\theta|^{-1/3})$ at parabolic infinity, zero limiting data do not eliminate this mode. A well-posed transversality theorem must instead fix the parabolic epoch or compare incoming and outgoing parabolic curves V_u, V_s on $z = 0$. Reversibility gives $V_s(\phi) = V_u(-\phi)$, so the target is $2V'_u(0) \neq 0$. Ordinary finite-cutoff calculations support this with a slope difference near -0.90623 , but no analytic or validated result is claimed.

There is a finite sufficient certificate for this transversality. Let $h'' = ah$, $h(0) = v_*$, $h'(0) = 0$, with $a = (4z^2 - r^2/2)/(z^2 + r^2/4)^{5/2}$. Fixed-softening comparison and an explicit early escape estimate give

$$\sqrt{8} < v_* < 4.$$

If $W = hz'' - h'z'$, then $W' = hbr'$ with $b = 3zr/[2(z^2 + r^2/4)^{5/2}]$. During the first binary contraction this proves $(h/z')' > 0$ as long as $h > 0$, and hence rules out a first zero of h . A force bound also gives

$$z(\pi/8) > \frac{\pi\sqrt{2}}{4} - \frac{2\pi^2}{27\sqrt{3}} > \frac{1}{2\sqrt{2}},$$

so $a > 0$ thereafter. Thus $h'(\pi/4) > 0$ forces h to grow forever.

This finite sign is now rigorous. Put $q = h/v$, $w = z'$, $p = q'$, and use eccentric anomaly $r = \cos^2 \psi$, $d\theta = r d\psi$. The regular compact system is

$$z_\psi = rw, \quad w_\psi = -\frac{2zr}{d^3}, \quad q_\psi = rp, \quad p_\psi = r\frac{4z^2 - r^2/2}{d^5}q, \quad (\text{CJ})$$

with $(z, w, q, p)(0) = (0, v, 1, 0)$ and terminal point $\psi = \pi/2$.

Theorem 10 (Computer-assisted finite Jacobi sign). *For every $v \in [14/5, 4]$, the solution of (CJ) exists through $\psi = \pi/2$ and*

$$p(\pi/2) > \frac{1}{125}.$$

In particular, the parabolic-separatrix field satisfies

$$h'(\pi/4) > \frac{14}{625} > 0.$$

Certificate description. CAPD 6.1.0 native outward-rounded intervals were used at the pinned commit 731079217a9254ea2948d742df2b170895effe7f. The interval $[14/5, 4]$ was divided into 256 overlapping slabs and each slab was propagated independently by an order-20 interval Taylor solver with a doubleton set. Every terminal interval had lower endpoint greater than $1/125$. The complete verifier prints exact hexadecimal binary64 endpoints; the dependency check, concise certificate, and symbolic regression test are included in the repository. Since $14/5 < \sqrt{8} < v_* < 4$ and $h = v_*q$, the stated corollary follows. $\square$

Theorem 11 (Restricted parabolic transversality). *The incoming and outgoing parabolic curves of the rectilinear equal-heavy-mass restricted problem cross transversely at the reversible apocenter-centered orbit.*

Proof. The collision-regularized stable-manifold implication can be checked directly. Compactify infinity by $z = 2/x^2$. In eccentric anomaly the field becomes

$$x_\psi = -\frac{r}{4}x^3w, \quad w_\psi = -\frac{r}{2}x^4(1 + r^2x^4/16)^{-3/2}.$$

It is analytic at $(x, w) = (0, 0)$ and through $r = 0$. Since $\int_0^\pi r d\psi = \pi/2$, its time- π map has, after $X = x$, $Y = w - \sqrt{2}x$, the form

$$X_1 = X - \frac{\pi}{8}X^3(Y + \sqrt{2}X) + O_7, \quad Y_1 = Y + \frac{\pi\sqrt{2}}{8}X^3Y + O_7. \quad (\text{PM})$$

Before this shear, $y = w/\sqrt{2}$ puts the exact map in the factored form

$$x_1 = x - kx^3(y + R_1), \quad y_1 = y - kx^3(x + R_2), \quad k = \frac{\pi\sqrt{2}}{8},$$

where analyticity of the exact flow and the x^3, x^4 factors in the vector field make R_1, R_2 analytic of order at least four (write $x = x_0 u$ to factor the exact flow by x_0^3 and x_0^4). The degree-four term obeys $p_1(X, 0) < 0$, $p_2(X, 0) = 0$, and $\partial_Y p_2(X, 0) > 0$ for $X > 0$. McGehee's theorem [9] therefore gives the fixed-phase parabolic stable curve $Y = \varphi(X)$ with $\varphi(0) = \varphi'(0) = 0$, analytic for $X > 0$. McGehee's graph construction supplies an arbitrarily small Lipschitz constant on a narrowed sector, hence $|\varphi'|$ is correspondingly small for $X > 0$.

Restricting (PM) to this graph gives, for positive constants c_i and small X ,

$$c_1 X^4 \leq X - F(X) \leq c_2 X^4, \quad 0 < F'(X) \leq 1 - c_3 X^3.$$

Thus $X_n^{-3} \asymp n$, $\sum X_n^3 = \infty$, and every stable-curve tangent $\delta X_n = (F^n)'(X_0)\delta X_0$ tends to zero.

Take both section speeds positive in the direction from $z < 0$ to $z > 0$. The reversor $\tilde{z}(t) = -z(-t)$ then gives $V_s(\phi) = V_u(-\phi)$. Tangency at zero would force both slopes to vanish. In extended variables $(\delta z, \delta w, \delta \phi)$, the global time-shift and pinned stable-phase tangents are explicitly

$$T_{\text{time}} = (v_*, 0, 1), \quad T_{\text{pin}} = (0, V'_s(0), 1) = (0, 0, 1).$$

Their difference is the fixed-phase stable tangent with z -component $h = z' - \eta_{\text{pin}}$. The validated theorem and $a > 0$ imply $h(\theta) \geq h(\pi/4) + (14/625)(\theta - \pi/4)$. But

$$\delta X_n = -\frac{X_n^3}{4}h(\theta_n), \quad \theta_n = \theta_0 + n\pi/2,$$

so $X_n^{-3} \asymp n$ makes $\liminf |\delta X_n| > 0$, contradicting tangent contraction. $\square$

Theorem 12 (Restricted phase windows). *There are one-sided open binary-phase intervals adjacent to the centered separatrix such that the incoming parabolic orbit escapes hyperbolically on one interval and turns at finite positive z on the other.*

Proof. The analytic section difference $\Delta(\phi) = V_u(\phi) - V_s(\phi)$ is odd and has $\Delta'(0) \neq 0$. It therefore has a strict linear sign on the two sides of zero. McGehee's stable curve separates the local hyperbolic and returning sectors. On the returning side $z'' < 0$ for $z > 0$; any trajectory that neither turns nor escapes with positive speed would be another parabolic point on the unique stable curve. Thus it turns in finite time. $\square$

The finite-turn side contains a singular sequence that prevents a uniform restricted no-brake argument.

Theorem 13 (Restricted turn resonances). *For every sufficiently large integer n , there is a finite-turn phase $\phi_n \rightarrow 0$ whose first outer turn is simultaneous with the n th heavy-binary apocenter. It is a nonsingular brake state of the collision-regularized restricted flow. If $Z_{t,n}$ and T_n are its turn height and time, then*

$$Z_{t,n} = (2n)^{2/3}(1 + o(1)), \quad T_n = \frac{\pi n}{2} + o(1).$$

Proof. At center phase ϕ , the binary mean anomaly is $\Lambda(\theta) = \phi + 4\theta$. The first-turn time $T(\phi)$ is continuous on the finite-turn side and tends to infinity at the transverse parabolic boundary; otherwise finite-time convergence would give a turn on the parabolic separatrix. Thus $\phi + 4T(\phi)$ crosses $2\pi n$ for every large n . For $Z = Z_t(\phi)$ and $w = z' > 0$, the exact comparison identity

$$\frac{d}{dz} \left(\frac{w^2}{2} - \frac{2}{z} \right) = \frac{2}{z^2} \left[1 - \left(1 + \frac{r^2}{4z^2} \right)^{-3/2} \right]$$

gives $T = (\pi/4)Z^{3/2}(1 + o(1))$. Combining this with $\phi_n + 4T_n = 2\pi n$ proves the asymptotics. $\square$

Intervening regularized heavy-heavy collisions mean these restricted brake states are not classical periodic counterexamples. For finite B , they form an infinite resonance sequence at which the full velocity residual must be split.

At a fixed resonance this splitting has an exact local form. If $a_n = z_{\theta\theta}(T_n) < 0$, $b_n = \partial_\phi z_\theta(T_n)$, and $\mathcal{A}(\phi) = \phi + 4T(\phi)$, the Jacobian of the two Jacobi radial-rate conditions with respect to phase and time is

$$D_n = Z_n(2b_n - a_n/2) = -\frac{Z_n a_n}{2} \mathcal{A}'(\phi_n).$$

Thus radial continuation by the implicit-function theorem is equivalent to simplicity of the turn-phase root. Simplicity is not required for local exclusion: uniform convergence of the divided transverse velocity makes any sequence of finite- B brakes converging to a restricted brake satisfy $0 = Z'_{B,x}/B \rightarrow C_n$. Its first coefficient is C_n/Z_n , where

$$C_n = -\frac{3}{4Z_n} \int_{-\infty}^{T_n} \frac{r^2 z p}{d^5} d\theta, \quad p'' = \frac{r^2 - 2z^2}{d^5} p, \quad p - z = O(|z|^{-1})$$

on the incoming tail. The first-order incoming theorem is now proved in smooth divided Levi-Civita normal variables. On the centered parabolic separatrix let p_- have the same incoming normalization and define

$$\mathcal{W}_\infty = \lim_{\theta \rightarrow +\infty} (z p'_- - z' p_-).$$

The exact identity

$$\frac{d}{d\theta} (z p'_- - z' p_-) = \frac{3r^2 z p_-}{2d^5}$$

makes this limit finite. Phase-uniform large-turn comparison, applied to $q = p/z$ and its Wronskian, proves

$$\boxed{Z_n C_n \longrightarrow -\mathcal{W}_\infty/2.}$$

Thus one scalar inequality $\mathcal{W}_\infty \neq 0$ implies $C_n \neq 0$ at every sufficiently late restricted resonance, uniformly over multiple roots. Ordinary finite-cutoff integrations give $\mathcal{W}_\infty \approx 0.4457$. A finite CAPD enclosure, joined to an analytic outgoing-Jost tail bound, proves

$$k_+(0) > 7/20, \quad k'_+(0) > 3/5, \quad \frac{11}{25} < \mathcal{W}_\infty = 2k_+(0)k'_+(0) < \frac{9}{20}.$$

Consequently $C_n < 0$ at every sufficiently late restricted resonance. The first-order tail theorem then proves that every fixed such resonance has a punctured finite- B neighborhood containing no full labelled brake. This fixed-resonance conclusion does not require radial simplicity. It is not uniform as $n \rightarrow \infty$, because $C_n \asymp Z_n^{-1} \rightarrow 0$, and it does not exclude a brake after later outer turns.

The joint scale is controlled by outer angular momentum. The outgoing restricted field has

$$p_- = \gamma z - \mathcal{W}_\infty \sqrt{z} + o(\sqrt{z}).$$

On the macroscopic radial Kepler arc $s = R \sin^2 \psi$, the collision-normalized transverse field $x = \sqrt{R} \sin \psi \cos \psi \sim \sqrt{s}$ solves $x'' = -2x/s^3$ and obeys $\dot{x}(\pi/2) = -1/R$. More invariantly, exact torque factorization and a collision-safe outgoing LC bootstrap prove, uniformly in first-turn height,

$$\frac{Y \times \dot{Y}}{B^{3/2}}(t_{\text{turn}}) \longrightarrow \frac{\mathcal{W}_\infty}{2} > \frac{21}{100}.$$

Hence a full labelled brake is impossible at the first outer turn throughout the accumulating skinny parabolic layer. The same tail estimate carries this positive normalized angular momentum back to a fixed large section on the inward leg. The next compact central encounter, and hence a brake after a later turn, remains uncontrolled.

One endpoint of that compact encounter map is already signed. Let (R_+, S_+) be the outgoing transverse Jost basis with $W(R_+, S_+) = -1$. Evenness and Wronskian preservation give the connection matrix

$$\begin{pmatrix} \gamma & (1 - \gamma^2)/\mathcal{W}_\infty \\ -\mathcal{W}_\infty & \gamma \end{pmatrix}.$$

The reflected returning field therefore has centered-parabolic outgoing Wronskian

$$\mathcal{C}_{2,\text{par}} = -2\gamma\mathcal{W}_\infty.$$

The extended pinned interval certificate proves $-1/100 < \gamma < -1/250$, hence $\mathcal{C}_{2,\text{par}} > 21/6250$. Thus a second-turn zero cannot emerge from the centered parabolic endpoint. The compact captured interior and its triple-collision transition remain open. On the equilateral restricted triple-collision ray the transverse coefficient is exactly $-1/(18\sigma^2)$, with indicial exponents $(3 \pm \sqrt{7})/6$; hence that endpoint requires fractional-power McGehee matching. More precisely, its remaining scalar target is

$$(\pi - \phi)^{(\sqrt{7}-1)/6} \mathcal{C}_2(\phi) \longrightarrow 4^{(\sqrt{7}-1)/6} A_- K_- > 0,$$

where A_- is the slow collision-mode coefficient of the returned Jost field and K_- is its universal first-turn Wronskian transfer.

One part of this endpoint problem is already analytic. In logarithmic time $\tau = -\log(1-s)$ and shape coordinate $y = Z/R$, the limiting base satisfies

$$y_{\tau\tau} - \frac{1}{3}y_\tau + \frac{2}{9}y[(y^2 + 1/4)^{-3/2} - 1] = 0.$$

Its increasing unstable branch from $y = -\sqrt{3}/2$ is unique modulo τ -translation, crosses $y = 0$ with $Z_s > 9^{1/3}\sqrt{5/18}$, and obeys $e^{-2\tau/3}y \rightarrow A_\infty > 0$. The last assertion follows from the strictly decreasing positive factor $e^{-2\tau/3}(y_\tau + y/3)$ and a tail contradiction. Thus the candidate universal light orbit stays separated from the prescribed binary collision. This does not yet prove that the finite-phase family selects that branch. The two scalar signs $K_- > 0$ and $A_- > 0$ are settled below.

There is also an analytic sign on the incoming half of the canonical slow transverse field. If $Q(y) = (1 - 2y^2)/(9(y^2 + 1/4)^{5/2})$ and $r = p_\tau/p$, then $r' = Q - r - r^2$ and $r \rightarrow -\alpha_-$ at the equilateral end. Tangent-line bounds for Q , together with $0 < y_\tau \leq y + \sqrt{3}/2$, make $g = -\alpha_- + (7/100)(y + \sqrt{3}/2)$ a strict lower barrier. Hence

$$p(0) > 0, \quad \frac{p_s(0)}{p(0)} \geq -\alpha_- + \frac{7\sqrt{3}}{200} > 0.$$

The passage through the prescribed binary collision and the subsequent short segment to the first turn can then be certified on a deliberately larger box. The energy identity gives $1/2 < y_\tau(0) < 3/4$, while the Riccati bound gives $p_s(0)/p(0) > 1/1000$. A pinned CAPD proof covers the center-speed interval in 256 dyadic slabs, crosses the binary collision in the analytic coordinate $\lambda = \text{sgn}(1-s)|1-s|^{1/3}$, and proves positive height and positive transverse derivative at the unique first turn. Therefore

$$K_- > 0.$$

The global coefficient is also positive, as proved next. The later transversality and shifted-Fuchsian argument then construct the selected local finite-phase branch.

The latter sign is nevertheless finite-dimensional. On the last binary infall, use the monotone separation r as independent variable and put $y = z/r$. The exact equations are

$$2r^2(1-r)y_{rr} + r(3-4r)y_r + y[(y^2 + 1/4)^{-3/2} - 1] = 0,$$

$$2r^2(1-r)p_{rr} - rp_r - \frac{1-2y^2}{2(y^2 + 1/4)^{5/2}}p = 0.$$

At equilateral collision their admissible shape exponent is $(-1 + \sqrt{19})/4$, while the transverse exponents are $\beta_\pm = (3 \pm \sqrt{7})/4$. Hence

$$\hat{A}_- = \lim_{r \downarrow 0} r^{-\beta_-} \frac{\beta_+ p - r p_r}{\beta_+ - \beta_-}$$

has the same sign as A_- . This reduces the global sign to one collision-stable/parabolic-manifold intersection and one scalar projection. A scalar comparison with the exact equilateral orbit shows that a branch with negative shape-stable coefficient remains in the cone $z < -(\sqrt{3}/2)r$ throughout the final infall. The global certificate below places every candidate parabolic connection in such a negative interval.

This match can be made square without adding a hidden dynamical parameter. On the regular section $r = 1/2$, let $U(\chi)$ be the incoming parabolic base curve, $C(a)$ the collision-stable base curve, $J(\chi)$ the returned Jost state, and $P_\pm(a)$ the canonical collision modes. The four residuals are

$$C(a) - U(\chi) = 0, \quad A_- P_-(a) + A_+ P_+(a) - J(\chi) = 0.$$

Their Jacobian is block lower triangular. In physical time the exact collision-mode Wronskian is

$$W(P_-, P_+) = -\sqrt{7},$$

so its determinant is $-\sqrt{7}$ times the oriented determinant of the two base curves. Thus only the base intersection requires a transversality certificate, after which

$$A_- = \frac{W(J, P_+)}{-\sqrt{7}}.$$

At incoming infinity, $z = -2/x^2$ and the regular fiber variables $P = x^2 p$, $\Omega = z\dot{p} - \dot{z}p$ tend directly to $(2\gamma, -\mathcal{W}_\infty)$ and obey an analytic system. Hence both singular ends attach to one compact interval-Newton problem. A backward finite-cutoff diagnostic gives $a \approx -0.24695$ and $\hat{A}_- \approx 0.31 > 0$.

Theorem 14 (Global triple-endpoint coefficient). *There is exactly one collision-stable/incoming-parabolic connection in the amplitude bracket*

$$-0.24696 \leq a \leq -0.24694$$

and it satisfies $A_- > 0$.

Proof. Analytic Fuchsian fixed-point estimates attach the full collision-stable family and its canonical fast mode at $r = 2^{-32}$. A pinned interval propagation carries the whole amplitude box to a large outgoing section and proves positive velocity along every accepted step. At the left endpoint the monopole energy $H_K = w^2/2 - 2/Z$ is positive; at the right endpoint $H_K + 1/(4Z^3) < 0$. The exact tail inequality

$$0 \leq H_K(\infty) - H_K(Z) \leq \frac{1}{4Z^3}$$

and connectedness therefore supply a parabolic member, without a uniqueness or transversality assumption.

The same propagation proves $Z > r/(2\sqrt{2})$ along the whole candidate box; the analytic collision cone and the large- Z tail cover the two omitted ends. Hence $f_Z > 0$ for the scalar light equation. For two amplitudes ordered by $b = -a$, their positive position and velocity differences obey $\Delta Z'' = c(t)\Delta Z$ with $c > 0$. The velocity difference is strictly increasing, so two different amplitudes cannot both have velocity tending to zero at infinity. This proves uniqueness within the bracket and orders the turn and escape sides.

On every parabolic member, a Volterra estimate in (P, Ω) attaches the exact returned Jost data $(2\gamma, -W_\infty)$ from infinity. Combining those boxes with the compact propagation gives the uniform rational margin

$$W(J, P_+) < -\frac{1}{2}.$$

Since $W(P_-, P_+) = -\sqrt{7}$, the exact projector gives

$$A_- = \frac{W(J, P_+)}{-\sqrt{7}} > 0.$$

The collision attachment is analytic; the compact enclosures use CAPD 6.1.0 at pinned commit 731079217a9254ea2948d742df2b170895effe7f. $\square$

Together with $K_- > 0$, this proves that the proposed limiting coefficient $4^{(\sqrt{7}-1)/6}A_-K_-$ is positive. It does not by itself prove the limit: one must still show that the finite-phase captured family selects this base connection and its normalized transverse field.

That singular local implication has an exact shifted-Fuchsian reduction. If ρ is the binary separation at center crossing and $\xi = -\log(r/\rho)$, then

$$2(1 - \rho e^{-\xi})y_{\xi\xi} + (-1 + 2\rho e^{-\xi})y_\xi + y[(y^2 + 1/4)^{-3/2} - 1] = 0.$$

The translated field converges to the universal hyperbolic saddle. A parameter-dependent inclination estimate selects its increasing unstable branch, while the fibered exchange estimate suppresses the fast mode by $\rho^{\sqrt{7}/2}$. Thus fixed-section convergence to the certified endpoint implies the full base-and-field asymptotic above. The remaining issue is the compact localization of any remotely continued captured component; the singular passage itself introduces no further sign ambiguity.

The certified connection is in fact a transverse base-curve intersection. Differentiating the collision contraction in the fixed terminal-collision gauge gives a tangent J with $J, J_s > 0$. The validated cone makes $J_{ss} = f_Z J > 0$, so its velocity component stays bounded away from zero. By contrast, the differentiable McGehee fixed-phase parabolic graph has contracting tangents, whose velocity components tend to zero at infinity. The section tangents therefore cannot be collinear.

It follows that one side of the incoming parabolic curve exits the translated saddle along the increasing universal peel. This produces a genuine local captured branch with center phase tending to π . On that branch the full fibered exchange theorem gives

$$(\pi - \phi)^{(\sqrt{7}-1)/6}\mathcal{C}_2(\phi) \longrightarrow 4^{(\sqrt{7}-1)/6}A_-K_- > 0.$$

Thus the restricted second-turn coefficient is positive on a punctured one-sided neighborhood of the certified triple endpoint. This is a collision-regularized restricted theorem. A finite-positive-mass wedge can now be transferred as follows.

Let $\omega_0(B), \omega_1(B) \rightarrow 0$ be respectively the fixed-section full-state and divided-transverse matching moduli supplied by the incoming-tail theorem. Put

$$\lambda_s = \frac{1 + \sqrt{19}}{6}, \quad \lambda_t = \frac{1 + \sqrt{7}}{6}, \quad \lambda_s > \lambda_t.$$

For an actual Pythagorean trajectory whose selected intercept $\Phi_{\text{ref}}(B)$ lies on the certified local restricted branch, let $\varepsilon = (\pi - \phi)/4$ be the comparison center-to-collision gap.

Theorem 15 (Finite-mass triple-endpoint wedge). *If $B_j \rightarrow 0$, $\varepsilon_j \rightarrow 0$, and*

$$\omega_0(B_j) + B_j = o(\varepsilon_j^{\lambda_s}),$$

then either the finite- B_j classical trajectory has a collision during the near-triple passage, or it has no full labelled brake near the continued restricted second turn. In the collision-free alternative, with $\eta_j = B_j \varepsilon_j^{-\lambda_t}$,

$$\frac{1}{\eta_j}(\mathcal{Z}_{j,x}, \mathcal{Z}_{j,x,s}) \longrightarrow -\frac{A_-}{2}(P_-, P_{-,s})$$

through the heavy-binary encounter and at the simple universal turn.

Proof. Homogeneity leaves the exact finite- B Jacobi field unchanged under $R, Z = \varepsilon^{2/3}(\mathcal{R}, \mathcal{Z})$ and $\theta = \theta_c + \varepsilon s$. If h is the signed unstable distance from the certified collision-stable graph, transversality gives $h \asymp \varepsilon^{\lambda_s}$, while the actual transverse displacement relative to the inner size is $\eta = B\varepsilon^{-\lambda_t}$.

The full homogeneous McGehee system, after zero-angular-momentum rotation reduction, has reduced log-size spectrum

$$-\frac{\sqrt{19}-1}{4}, \quad \frac{1+\sqrt{19}}{4}, \quad 1 - \frac{3-\sqrt{7}}{4}, \quad 1 - \frac{3+\sqrt{7}}{4}.$$

A simultaneous longitudinal/transverse graph-transform bootstrap bounds the translated error by

$$O\left(\frac{\omega_0 + B}{h} + B|\log h| + \eta + \omega_1\right) = o(1).$$

It also proves that the actual and restricted center scales have ratio $1 + o(1)$. At the heavy-binary encounter, reflection and Hadamard division of the smooth Levi-Civita field leave a direct source of size $B/\eta = \varepsilon^{\lambda_t} \rightarrow 0$. Thus the divided outer state passes through the regularized block and converges in physical C^1 near the separated simple turn. Since $p = -2\xi$, its limit is the one displayed. Finally $A_- > 0$ and $K_- = Z_*(s_t)P_{-,s}(s_t) > 0$, so the limiting transverse velocity is nonzero. A true finite- B collision instead terminates the classical solution. Full details, including the fixed-inertial reconstruction, are in `docs/FINITE_B_TRIPLE_ENDPOINT.md`. $\square$

In particular, with

$$\Delta(B) = (\omega_0(B) + B)^{3/(1+\sqrt{19})},$$

all sufficiently small finite masses are excluded from a brake on the local branch for $\Delta(B) \leq \varepsilon \leq \varepsilon_0$. The complementary boundary layer $0 < \varepsilon < \Delta(B)$ remains open.

There is nevertheless no brake *inside* that boundary layer's near-triple tube. Indeed the heavy pair has physical separation $X = BR$, while at every full brake energy conservation requires $U = U_0 = AB + (AB)^{-1}$. Positivity of the other pair terms gives the exact strict condition

$$|R| > \frac{A}{BU_0} = \frac{1 - B^2}{1 + B^2 - B^4} = 1 - 2B^2 + O(B^4).$$

Thus every state with $|R| = o(1)$ is separated from the brake boundary. The unresolved boundary layer concerns the signed outgoing data and possible later motion after the heavy pair re-expands, not a brake during the close passage.

Theorem 16 (Joint endpoint reduction). *Let $B_j \downarrow 0$ be collision-free finite-mass states whose selected restricted intercepts approach the certified triple endpoint. There are intrinsic longitudinal and reflection-odd unstable amplitudes H_B, T_B for which*

$$\frac{T_B}{B} \rightarrow -\frac{\hat{A}_-}{2} \neq 0.$$

Put

$$\mu = \frac{1 + \sqrt{19}}{4}, \quad \tau = \frac{1 + \sqrt{7}}{4}, \quad \rho_T = |T_B|^{1/\tau}, \quad \kappa_B = \frac{H_B}{|T_B|^{\mu/\tau}}.$$

If $\kappa_{B_j} \rightarrow \kappa \in \mathbb{R}$, scaling size by ρ_T and time by $\rho_T^{3/2}$ gives the one-parameter planar massless restricted limit

$$\mathcal{R}_{ss} = -2\Phi(\mathcal{R}), \quad \mathcal{Z}_{ss} = -\Phi(\mathcal{Z} + \mathcal{R}/2) - \Phi(\mathcal{Z} - \mathcal{R}/2).$$

If $\kappa_{B_j} \rightarrow \pm\infty$, longitudinal scaling instead selects the corresponding rectilinear exit. A genuine finite- B collision is always the classical-termination alternative.

Proof. Use the parity-adapted nonlinear unstable amplitudes of the persisted McGehee rest point. In the auxiliary family with equal heavy masses $(1, B, 1)$, heavy exchange and reflection split the stable foliation and spectral projectors into exact even and odd blocks. Since $A(B) = \sqrt{1 - B^2} = 1 + O(B^2)$, the Pythagorean field differs from this auxiliary field by $O(B^2)$ in C^2 . Subtracting the moving stable graph therefore does not alter the divided first-order odd coefficient, and the proved Fuchsian matching gives $T_B/B \rightarrow -\hat{A}_-/2$.

After translations, $\text{SO}(2)$, log-size/time, and fixed energy have been removed, the unstable space has exactly these two scalar modes. The longitudinal stable mode, fast odd mode, and energy/clock variables lie in the stable or triangular block; no planar angle remains after zero-angular-momentum reconstruction.

The two unstable exponents satisfy $1 < \mu/\tau < 2$. Thus $H \mapsto H + O(T^2)$ changes κ_B by $o(1)$, while $B|\log B| \rightarrow 0$ absorbs finite- B exponent drift over the saddle dwell. The exponential dichotomy gives, on every fixed translated interval, the uniform error

$$O_L\left(|\kappa_B - \kappa| + B|\log |T_B|| + B + |T_B|^{2-\mu/\tau} + e^{-c|\log |T_B||}\right) + o_L(1).$$

Translation by $-\log \rho_T$ and Newtonian homogeneity therefore give the displayed one-parameter limit. Across a separated limiting heavy-binary collision the convergence is only in the common Levi-Civita variables and clock; an exact collision of a finite member is not continued. $\square$

This theorem removes the arbitrary two-parameter character of the shrinking boundary layer, but it does not yet exclude a later brake. The remaining local task is a global classification and a nonvanishing scattering theorem for the single real family indexed by κ .

Lemma 17 (Planar shape identities). *On a zero-energy radial-binary half put*

$$R = 9^{1/3}t^{2/3}e_x, \quad Z = 9^{1/3}t^{2/3}w(\zeta), \quad \zeta = \log t.$$

Then

$$w_\zeta \zeta + \frac{1}{3}w_\zeta = \nabla W(w),$$

where

$$W = \frac{1}{9} \left(|w|^2 + |w + e_x/2|^{-1} + |w - e_x/2|^{-1} \right).$$

If $\mathcal{E} = |w_\zeta|^2/2 - W$ and $\Lambda = w \times w_\zeta$, then

$$\mathcal{E}_\zeta = -\frac{|w_\zeta|^2}{3}, \quad \Lambda_\zeta + \frac{\Lambda}{3} = -xyK(x, y), \quad K > 0.$$

The sharp global bound is $W \geq 11/36$, with equality only at the two equilateral shapes. Hence a nonhomothetic equilateral-stable orbit has $w_\zeta \neq 0$ at every finite point and has no compact collision-free backward-captured component. Moreover, on $|y| \geq \sqrt{3}/2$,

$$W_{xx} \geq \frac{1}{9} \left(\frac{7}{4} - \frac{8}{3\sqrt{3}} \right) > 0.$$

Proof. The first two identities follow by direct substitution and scalar/cross products. If $d_\pm = |w \pm e_x/2|$, then

$$9W = \sum_{\pm} \left(\frac{d_\pm^2}{2} + \frac{1}{d_\pm} \right) - \frac{1}{4} \geq \frac{11}{4},$$

since $d^2/2 + d^{-1} - 3/2 = (d-1)^2(d+2)/(2d)$. The strict energy identity excludes a finite shape rest. A compact backward tail would have an invariant α -limit set on which the energy is constant, hence $w_\zeta = 0$ and $\nabla W = 0$; its energy would be at most $-11/36$, contrary to the strict stable-tail energy.

For the last inequality, set $Y = y^2$ and $g_Y(q) = (Y - 2q)/(Y + q)^{5/2}$. Then

$$9W_{xx} = 2 - g_Y((x+1/2)^2) - g_Y((x-1/2)^2).$$

One squared argument is at least $1/4$; monotonicity of g_Y gives an upper bound $Y^{-3/2} + (Y - 1/2)/(Y + 1/4)^{5/2}$, whose maximum for $Y \geq 3/4$ is $8/(3\sqrt{3}) + 1/4$. $\square$

The angular identity fixes a nonzero sign on every stable tail that remains in the outer strip. It is not yet global: at $\kappa = 0$ the stable manifold bends into the inner strip with the exact coefficient $(2\sqrt{21} - 5\sqrt{3})/4 > 0$. Thus the next problem is a finite transition classification, not an invariant-strip assertion. On the incoming half $t = t_c - \theta$, so the physical angular-momentum sign is opposite the ζ -oriented sign above.

For the collision shoot we use the exact quartic stable jet. With $p = Te^{-a_T\zeta}$ and $\nu = He^{-a_L\zeta}$ it has the normalized form

$$x = p + Ap\nu + Bp^3 + Cp\nu^2 + O_4, \quad q = \nu + Kp^2 + D\nu^2 + Ep^2\nu + F\nu^3 + O_4,$$

Its degree-four correction is

$$x_{[4]} = Gp^3\nu + Hp\nu^3, \quad q_{[4]} = Ip^4 + Jp^2\nu^2 + L\nu^4.$$

All twelve coefficients are explicit elements of $\mathbb{Q}(\sqrt{3}, \sqrt{7}, \sqrt{19})$. Direct substitution kills the full residual through total degree four.

Theorem 18 (Stable-tail enclosure). *Let P_4 be this quartic parameterization and equip vector power series with the weighted coefficient norm*

$$\|F\|_{r_T, r_L} = \sum_{i,j} (|F_{ij}^x| + |F_{ij}^q|) r_T^i r_L^j.$$

On $|p| \leq 1/200$, $|\nu| \leq 1/2500$, there is a unique normalized analytic stable parameterization $P = P_4 + E$, absolutely convergent on the closed polydisc, such that

$$\|E\|_{1/200, 1/2500} < \frac{1}{125000000}.$$

On the half polydisc,

$$\|E\| < \frac{1}{4000000000}, \quad \|\mathcal{D}E\| < \frac{1}{800000000},$$

and

$$\|\partial_\kappa E\| < 10^{-9}, \quad \|\partial_\kappa \mathcal{D}E\| < 5 \cdot 10^{-9}.$$

These half-radius estimates apply to the real collision-shooting tail $p = -e^{-a_T \zeta}$, $\nu = \kappa e^{-a_L \zeta}$ for $5/4 \leq \kappa \leq 13/10$ and $\zeta \geq 10$.

Proof. The homological inverse on degrees at least two has norm below $30/17$ because its smallest divisor is $(5+2\sqrt{7})/18 > 17/30$. Exact algebraic coefficient bounds on P_4 and an eightyfold Cauchy dilation give residual norm $< 13/80^5$. On the base-radius ball $R = 271/50000$, a rational majorant gives $\|D^2 \nabla W\| < 9$, hence contraction factor $(30/17)9R < 1/10$. The correction is consequently below $13/1671168000 < 1/125000000$. Since it begins in degree five, restriction to half radii and coefficient differentiation give the displayed state and parameter-tangent bounds. Exact Taylor lower bounds for the two exponentials put the stated real tail inside the half polydisc. $\square$

The degree-five defect can also be removed exactly. Reflection symmetry gives

$$E_5^x = Mp^5 + Np^3\nu^2 + Op\nu^4, \quad E_5^q = Pp^4\nu + Qp^2\nu^3 + R\nu^5,$$

where all six coefficients lie in $\mathbb{Q}(\sqrt{7}, \sqrt{19})$ and have signs $(+, -, +, -, +, -)$. Exact homological substitution kills every degree-five residual. Since homogeneous degrees are disjoint in the coefficient ℓ^1 norm used above, writing $P = P_4 + E_5 + E_{\geq 6}$ preserves the four norm bounds while restriction to normalized radius λ gains λ^6 instead of λ^5 . At $\zeta = 15$ the collision verifier has $\lambda < 0.04403$. The six radical expressions and exact symbolic regression are recorded in the companion quintic-enclosure note.

Lemma 19 (Forced light-collision chart). *Let $\epsilon = \pm 1$ select the heavy primary at $\epsilon R/2$ and put $q = Z - \epsilon R/2$. On either heavy-binary half let $t = |\theta - t_c|$ and $\eta = dt/d\theta = \pm 1$, where θ is physical time. Then derivatives of order two in t and θ agree, and*

$$q_{tt} = -\Phi(q) + G_\epsilon(q, t), \quad G_\epsilon = \epsilon\Phi(R) - \Phi(q + \epsilon R).$$

With complex Levi-Civita variables

$$q = u^2, \quad dt = |u|^2 d\sigma, \quad h = \frac{|q_t|^2}{2} - \frac{1}{|q|},$$

the analytic regular field is

$$\begin{aligned} u_\sigma &= v, \\ v_\sigma &= \frac{h}{2}u + \frac{|u|^2\bar{u}}{2}G_\epsilon, \\ h_\sigma &= 2\operatorname{Re}(uv\overline{G_\epsilon}), \\ t_\sigma &= |u|^2. \end{aligned}$$

It preserves $2|v|^2 - 1 - h|u|^2 = 0$ and hence $|v| = 1/\sqrt{2}$ at collision. The real-analytic chart domain is $t > 0$, $u^2 + \epsilon R(t) \neq 0$; the latter inequality excludes the other primary. At collision $t_\sigma = t_{\sigma\sigma} = 0$ and $t_{\sigma\sigma\sigma} = 1$, so the radial clock is monotone through the regularized collision. Away from collision the physical velocity is

$$Z_\theta = \eta \left(\frac{2v}{\bar{u}} + \frac{\epsilon}{2}R_t \right).$$

Thus increasing regularized time has the opposite physical orientation on the incoming half.

Consequently an isolated collision in the projective family is a zero of the square map

$$(\kappa, \sigma) \longmapsto (\operatorname{Re} u, \operatorname{Im} u).$$

Here the asymptotic normalization of κ , one LC deck lift, and the origin and orientation of σ are fixed. Moreover the selected-primary relative angular momentum is exactly

$$\ell = q \times q_t = 2 \det(u, v), \quad \partial_\kappa \ell|_{u=0} = 2 \det(\partial_\kappa u, v).$$

Thus the interval-Newton determinant is precisely half the derivative of a signed physical impact parameter (with the branch factor η when physical time is restored). With $dt/d\sigma = |u|^2$ fixed it is invariant under a κ -dependent LC-time translation and under a deck change; its sign still depends on the orientation of κ . If $D_c = \det(\partial_\kappa u, v) \neq 0$ at the zero, the equation $u \cdot v = 0$ defines a unique nearby local closest-passage time. Orthogonal projection onto $v_c^\perp$ and $|v_c|^2 = 1/2$ give

$$r_{\min, \text{local}} = 2D_c^2(\kappa - \kappa_c)^2 + O(|\kappa - \kappa_c|^3), \quad \ell = 2D_c(\kappa - \kappa_c) + O(|\kappa - \kappa_c|^2).$$

This proves local avoidance on both punctured sides of a transverse zero, without asserting absence of other collisions. This was the interval-Newton target closed below. The resulting transverse zero separates nearby continuations locally; identifying full adjacent collision-free components still requires excluding every remaining collision zero. Its regularized continuation is not a physical continuation of a classical finite-mass collision.

Validated finite stage. For $1.26793 \leq \kappa \leq 1.26794$, pinned CAPD 6.1.0 native-interval integration propagates the rigorous stable-tail box from $\zeta = 10$ to $\zeta = 0.3$, converts it to the selected-primary LC chart, and propagates the reversed LC flow for the exact duration 1071249/1000000. Every accepted LC step satisfies $t > 1$, $R > 2$, and distance to the other primary greater than one. At the final fixed time,

$$-0.017807 < \operatorname{Re} u < 0.016661, \quad -0.003953 < \operatorname{Im} u < 0.004019.$$

This validates finite propagation and chart safety, but not a collision zero: the present interval coordinate conversion loses the affine κ correlation, so the terminal box is too wide for interval Newton.

Theorem 20 (Transverse planar collision boundary). *The normalized lower-equilateral planar family with $T = -1$ has exactly one selected-primary collision zero in*

$$\begin{aligned} 1.2679350755 &\leq \kappa \leq 1.2679352755, \\ 1.0712484057 &\leq -\sigma \leq 1.0712486057. \end{aligned}$$

It lies in

$$1.2679351752 < \kappa_c < 1.2679351759, \quad -1.0712485063 < \sigma_c < -1.0712485051,$$

and is transverse, with

$$-3.52212 < \det(\partial_\kappa u, v) < -3.38401.$$

Proof. At $\zeta = 15$, restriction of the analytic tail enclosure gains the factor λ^5 , where

$$\lambda = \max\{400e^{-15a_T}, 6500e^{-15a_L}\} < 0.04403,$$

because the state and parameter-tangent corrections all begin in total degree five. A pinned CAPD C^1 doubleton is retained through three fields: reversed shape flow to $\zeta = 0.3$, a unit construction flow writing the exact nonlinear LC lift, and reversed LC flow. Whole-step enclosures keep both shape primary distances positive, prove $y > 0$ and $u_r, u_i > 0$ at the selected lift, and retain $t > 1$, $R > 2$, and other-primary distance greater than one.

For $S = -\sigma$, the computed Jacobian columns are $(\partial_\kappa u, -v)$. Its determinant lies in

$$[3.3840110561, 3.5221124176].$$

Rigorous interval Gaussian elimination puts the interval Newton image strictly inside the displayed parameter-time box, proving existence and uniqueness there. Restoring increasing σ gives the stated determinant sign. The hexadecimal certificate and pinned replay are recorded in the companion computer-assisted proof note. $\square$

The adjacent planar component contains a second collision boundary, involving the opposite primary.

Theorem 21 (Second planar collision boundary). *The same normalized planar family has exactly one opposite-primary collision zero in*

$$\begin{aligned} 1.2640119181 &\leq \kappa \leq 1.2640119321, \\ 1.0275748704 &\leq S_2 \leq 1.0275749704. \end{aligned}$$

It lies in

$$1.2640119191 < \kappa_2 < 1.2640119311, \quad 1.0275748796 < S_2 < 1.0275749612,$$

is reached without an earlier collision, and is transverse. For the fixed second-chart clock, the interval Jacobian determinant lies in

$$98.8377 < \det D_{(\kappa, S_2)} F_2 < 99.2930.$$

Proof. Propagate the stable-tail enclosure in the positive-primary LC chart for the exact duration $7/4$. Whole-step intervals give $t > 1$, $R > 2$, distance to the opposite primary greater than one, and selected-primary distance greater than 10^{-8} . Thus this leg is classically collision-free. At its fixed endpoint make the exact chart change

$$q_- = q + Re_x, \quad \dot{q}_- = \frac{2uv}{|u|^2} + \frac{2R}{3t}e_x, \quad u_-^2 = q_-.$$

The switch enclosure proves $\text{Im } q_- < 0$, and its image fixes the exact lift by $\text{Re } u_- > 1$, $\text{Im } u_- < 0$. In the resulting forced LC chart the positive primary stays at squared distance greater than $1/100$, $t > 1/2$, and $\text{Re } v_- > 1/2$ through the candidate collision. Since $(\text{Re } u_-)_S = -\text{Re } v_- < 0$ and $\text{Re } u_- > 1$ at entry, the terminal zero is the first selected- pair collision rather than a regularized recollision. A second C^1 doubleton therefore encloses the square residual $F_2 = (\text{Re } u_-, \text{Im } u_-)$. Pinned interval Newton maps the displayed box strictly into

$$[1.264011919130, 1.264011931053] \times [1.027574879669, 1.027574961151].$$

The displayed determinant interval excludes zero, proving existence, uniqueness in the box, and transversality. The collision time satisfies $0.6152 < t < 0.6153$, so it precedes the prescribed heavy-binary collision. The hexadecimal certificate is recorded in the second-collision proof note. $\square$

Theorem 22 (Second-boundary collision-or-escape box). *For every limiting planar parameter*

$$1.2640099161 \leq \kappa \leq 1.2640126461,$$

the classical branch either collides or, if it remains collision-free, reaches a strict hyperbolic-escape cone. The regularized continuation of the exact collision root reaches that cone. Every sufficiently small compatible positive-light-mass member near this box likewise either collides classically or escapes, and hence has no second brake.

Proof. Continue the negative-primary LC box beyond the preceding root enclosure to a fixed separated section and set

$$q_+ = q_- - Re_x, \quad \dot{q}_+ = \frac{2u_-v_-}{|u_-|^2} - \frac{2R}{3t}e_x.$$

The interval image proves $\text{Im } q_+ < 0$. On that sheet use the cancellation-free exact lift

$$u_{+,i} = -\sqrt{\frac{|q_+| - q_{+,x}}{2}}, \quad u_{+,r} = \frac{q_{+,y}}{2u_{+,i}}, \quad v_+ = \frac{\overline{u_+}\dot{q}_+}{2}.$$

The resulting forced positive-primary LC field has nonsingular forcing

$$g_+ = \frac{e_x}{R^2} - \frac{q_+ + Re_x}{|q_+ + Re_x|^3}.$$

Pinned CAPD carries 195 abutting root-width tiles through this third chart and then through the signed- $\lambda = \sqrt[3]{t}$ heavy-binary bridge. The tile centers are $1.2640119251 + 14j \cdot 10^{-9}$, $-143 \leq j \leq 51$, and every radius is 7×10^{-9} ; their outward-rounded endpoints cover the displayed interval. The squared distances to the unregularized primary in the two post-root LC blocks, and to either light primary in the bridge, are respectively greater than

$$0.1760, \quad 0.1264, \quad 0.004936.$$

At the terminal section the restricted escape inequality has margin greater than 2.7591, while the hierarchical finite-mass inequality has margin greater than 2.8882. Therefore any selected-pair zero is a terminating classical collision; every member without such a zero agrees with the regularized flow and escapes.

All preterminal maps form a finite analytic chain of pair-LC charts and a heavy-pair LC chart, with every nonselected denominator uniformly separated. Thus the Newtonian-scaled terminal state and inner energy converge for small positive light mass. The branch-independent hierarchical-energy lemma absorbs the full future $O(B)$ tidal work, transferring the strict second margin. This proves the positive-mass dichotomy without physically continuing any collision. The hexadecimal enclosure and trusted computing base are recorded in the companion second-boundary escape note. $\square$

Theorem 23 (Third and fourth planar collision boundaries). *Immediately below the preceding collision-or-escape box, the limiting planar family has two further classically reached transverse collision roots. The positive-primary root satisfies*

$$\begin{aligned} 1.2640090993171269 &\leq \kappa_3 \leq 1.2640090999643190, \\ 0.5884119895288813 &\leq S_3 \leq 0.5884197454438135, \end{aligned}$$

and

$$5224.5883 < \det D_{(\kappa, S_3)} F_3 < 5227.2175.$$

The subsequent negative-primary root satisfies

$$\begin{aligned} 1.2640090985775962 &\leq \kappa_4 \leq 1.2640090992806481, \\ 0.3710663877519410 &\leq S_4 \leq 0.3726139177003617, \end{aligned}$$

and

$$417669.25 < \det D_{(\kappa, S_4)} F_4 < 429486.31.$$

The exact member at the fourth root avoids the preceding positive-primary collision: outward-rounded interval subtraction gives a separation greater than 1.2058×10^{-11} between the corresponding root projections. The ordinal labels refer to the successive encounter charts; uniqueness is claimed only in the displayed boxes.

Proof. Continue the negative-primary LC chart from the common incoming section and make the exact positive-primary switch used in the preceding theorem. In reversed regularized time, put

$$F_3(\kappa, S_3) = (\operatorname{Re} u_+, \operatorname{Im} u_+), \quad \partial_{S_3} u_+ = -v_+.$$

Pinned interval Newton maps the input box

$$[1.2640090992, 1.2640091000] \times [0.588396, 0.588436]$$

strictly into the first displayed enclosure. Every earlier selected distance is positive, the nonselected primary stays separated, and $\operatorname{Im} v_+ < -1/2$ throughout the positive-primary chart. Hence $\operatorname{Im} u_+$ increases strictly and the enclosed zero is the first selected collision.

For the next passage reconstruct

$$\lambda = t^{1/3}, \quad R = 9^{1/3} \lambda^2, \quad Z = q_+ + \frac{R}{2} e_x, \quad V = \dot{q}_+ + \frac{R}{3t} e_x,$$

propagate the signed- λ bridge for the exact duration 97/5000, and set

$$q_- = Z + \frac{R}{2}e_x, \quad \dot{q}_- = V + \frac{9^{1/3}}{3\lambda}e_x.$$

On the certified sheet $q_{-,x} > 0.05$, the lift

$$u_{-,r} = \sqrt{\frac{|q_-| + q_{-,x}}{2}}, \quad u_{-,i} = \frac{q_{-,y}}{2u_{-,r}}, \quad v_- = \frac{\overline{u_-}\dot{q}_-}{2}$$

is analytic even across $q_{-,y} = 0$. With $F_4 = (\text{Re } u_-, \text{Im } u_-)$, interval Newton maps

$$[1.2640090985, 1.26400909929] \times [0.37084, 0.37284]$$

strictly into the second displayed enclosure. The other primary remains at squared distance greater than $1/10$, and $\text{Re } v_- > 1/2$ makes $\text{Re } u_-$ decrease strictly to its first zero.

Finally, a separate interval-Newton replay on

$$[1.2640090981, 1.2640091001] \times [0.588316, 0.588516]$$

contains every possible intervening positive-primary zero and proves that its unique root has

$$1.2640090992927064 \leq \kappa_3 \leq 1.2640090999346494.$$

Subtracting the fourth-root projection with outward rounding gives the stated positive gap. The positive chart stays past its zero sheet thereafter, while all bridge and fourth-chart separations are strict. Thus the fourth root is reached classically without the third collision. The pinned hexadecimal enclosures and complete chart-domain audit appear in the companion proof note. $\square$

Theorem 24 (Outgoing entry after the fourth boundary). *For every*

$$1.2640090985 \leq \kappa \leq 1.26400909929,$$

the limiting classical branch either collides or reaches a uniformly analytic fifth positive-primary LC chart. Along the regularized common continuation, the intervening negative-chart other-primary squared distance is greater than 0.333865, while at the switch

$$|u_-|^2 > 0.213057, \quad q_{+,x} < -0.577809, \quad |q_+|^2 > 0.333864.$$

No terminal escape assertion is made.

Proof. Evaluate the fourth negative-primary chart at the fixed reversed clock $S_4 = 0.37284$, which is strictly beyond the preceding root-clock enclosure, and continue for the exact additional duration 0.42716. Pinned whole-step enclosures give the displayed separation bounds and keep $t > 0.05$. The exact switch

$$q_+ = u_-^2 - Re_x, \quad \dot{q}_+ = \frac{2u_-v_-}{|u_-|^2} - \frac{2R}{3t}e_x$$

is therefore nonsingular. Since $q_{+,x} < -0.05$, the sheet

$$u_{+,i} = -\sqrt{\frac{|q_+| - q_{+,x}}{2}}, \quad u_{+,r} = \frac{q_{+,y}}{2u_{+,i}}$$

is uniformly analytic; the interval image has $u_{+,i} < -0.760137$ and $t > 0.240640$. A selected-pair zero anywhere in the chain is a terminating classical collision. Conditional on no such zero, classical uniqueness identifies the trajectory with the regularized chain and proves arrival at the fifth chart. The pinned hexadecimal record appears in the companion proof note. $\square$

Theorem 25 (Outgoing section after the fifth passage). *For every*

$$1.2640090985 \leq \kappa \leq 1.26400909929,$$

the limiting classical branch either collides or crosses the outgoing fifth positive-primary LC section $c_i = 1/20$. At the crossing,

$$|c|^2 > 0.002166715533586436, \quad r_{\text{other}}^2 > 0.4599473198119462, \quad -d_i > 0.740153775124572.$$

In particular the section is collision-free and uniformly transverse. No terminal escape assertion is made.

Proof. Use the exact quintic tail enclosure at $\zeta = 15$ and retain one CAPD doubleton from the stable tail. Dimension-preserving nonlinear chart maps act directly on this set. For a dimension-changing map $y = F(x)$, append dormant coordinates and integrate the exact auxiliary graph system

$$x' = 0, \quad y' = F(x), \quad y(0) = 0$$

for one auxiliary unit. This preserves all correlation generators through dimensions $13 \rightarrow 19 \rightarrow 24 \rightarrow 30$ instead of replacing each nonlinear image by an interval product. The final 30-dimensional system freezes the old coordinates and propagates the fifth six-dimensional LC state to the Poincare section. The pinned interval image gives the three displayed strict bounds. Every ordinary denominator and square-root sheet in the preceding chain stays inside its analytic domain. A selected-pair zero is a terminating classical collision; conditional on no such zero, uniqueness identifies the regularized and classical branches up to the section. Auxiliary graph time is bookkeeping and is not physical flight time. $\square$

The next two close passages overlap, so a single-centre chart is insufficient. They admit one simultaneous polynomial regularization. Write

$$s = \sin(\xi/2), \quad c = \cos(\xi/2), \quad s^2 + c^2 = 1, \quad z = \frac{c^2 - s^2}{2}, \quad A = -sc, \quad g = |A|^2.$$

Then the two primary distances are $|s|^2$ and $|c|^2$, so g is their product. If

$$E = \frac{|z_\zeta|^2}{2} - W(z), \quad p = g\xi_\zeta, \quad qquad \frac{d\zeta}{d\sigma} = \eta g,$$

put

$$\begin{aligned} K_0 &= \frac{2}{9}Az(\bar{A}^2 - \bar{z}^2), \\ K_{\text{pair}} &= \frac{s\bar{c} - c\bar{s}}{9}, \\ \mathcal{P} &= -2A\bar{z}E + K_0 + K_{\text{pair}} - \frac{g}{3}p. \end{aligned}$$

The simultaneous two-centre field is

$$s_\sigma = \eta cp/2, \quad c_\sigma = -\eta sp/2, \quad p_\sigma = \eta \mathcal{P}, \quad E_\sigma = -\eta |p|^2/3, \quad \zeta_\sigma = \eta g.$$

It is polynomial in real and imaginary parts. It preserves $s^2 + c^2 = 1$ and

$$|p|^2 - 2gE - \frac{2}{9}(g|z|^2 + |s|^2 + |c|^2) = 0.$$

At either collision the latter gives $|p|^2 = 2/9$. From the fifth positive LC state $q_+ = u^2$ it is initialized algebraically by $s = iu/\sqrt{R}$, $c = \sqrt{1 - s^2}$. Thus the remaining obstruction is a finite validated propagation in a regular field, not the existence of a common analytic chart.

Theorem 26 (Fourth-boundary collision–ejection classification). *The fourth limiting collision root has the refined enclosure*

$$\begin{aligned} 1.26400909893331527 &\leq \kappa_4 \leq 1.26400909893331813, \\ 0.371840180067887265 &\leq \tau_4 \leq 0.371840186320052546, \end{aligned}$$

and its transverse negative-primary LC velocity satisfies

$$-3.832349985449157 \cdot 10^{-7} \leq v_i \leq -3.681944345249734 \cdot 10^{-7} < 0. \quad (38)$$

On the regularized collision–ejection continuation, the following positive and negative two-centre foci are both noncollisional. Their normalized selected distances obey

$$r_+ \geq 3.964960196340643 \cdot 10^{-11}, \quad r_- \geq 1.098165821369775 \cdot 10^{-6}. \quad (39)$$

At the next positive focus the selected distance exceeds 0.0506366673765623. The continuation then reaches the strict restricted escape cone with margin greater than 10.2599662232953, without another light–primary collision; the finite-mass-form comparison expression has margin greater than 10.3953509156741.

Proof. The degree-eight stable graph leaves a rigorously bounded order-nine tail at the entrance section. A 200-bit MPFR-backed CAPD integration and interval Newton step first maps the fourth-root box strictly into

$$[1.26400909893313121, 1.26400909893350310]$$

in the parameter coordinate. A nested Newton localization of the same unique root gives the displayed parameter and clock boxes. Re-evaluation on that image gives (38). The nesting is only a refinement: existence and uniqueness come from the first strict inclusion.

At the negative collision introduce elliptic two-centre coordinates

$$\alpha = \pi, \quad \beta = 0, \quad p_r = -\frac{2v_r}{3}, \quad p_i = -\frac{2v_i}{3}.$$

The first positive focus is $\alpha = 2\pi$, where $r_+ = \sinh^2(\beta/2)$. An outward-widened rational hull of the independent multiprecision certificate propagates under the simultaneous regularized field to

$$-1.310804394749059 \cdot 10^{-5} \leq \beta \leq -1.259358598132686 \cdot 10^{-5},$$

which proves the first inequality in (39). The next negative focus has

$$-0.002181489558789266 \leq \beta \leq -0.002095867762413745,$$

and hence the second inequality in (39).

At the next positive focus, embed the exact two-centre-to-ordinary map as an auxiliary graph ODE instead of taking a componentwise interval hull. This retains all six incoming doubleton generators. The resulting ordinary heavy-binary bridge crosses the regularized binary section and reaches the terminal escape section with both light–primary squared distances greater than 0.00262385439850217. Direct outward-rounded evaluation gives the two strict escape margins stated above. The pinned source versions, hexadecimal input boxes, and reproduction commands are recorded in the companion computer-assisted proof note. $\square$

The theorem classifies the mathematical collision–ejection continuation of the boundary member. Its classical Newtonian solution itself ends at the fourth collision; no postcollision trajectory is being declared physical.

Corollary 27 (Local limiting collision or escape). *There is an open parameter interval containing κ_4 such that every member of the limiting planar family either suffers a terminating classical collision or, if it remains collision-free, enters the strict escape cone. In particular no collision-free member in this interval is periodic.*

Proof. Take the finite regularized chart chain from a compact incoming section to the terminal escape section. All nonselected distances and all chart sheets have strict margins on the certified boundary chain. The fourth collision is transverse by (38), so the regularized flow and the collision sheet depend analytically on the initial parameter near the root. Compactness gives a single neighborhood on which the chart chain is defined until either a selected collision occurs or the terminal section is reached. In the latter case the strict clearance and escape inequalities persist by continuity. The former case terminates the classical orbit, while the latter excludes a second brake and every bounded periodic orbit. $\square$

Theorem 28 (Explicit common-clock continuum). *Every member of the limiting planar family with*

$$1.264009099014 \leq \kappa \leq 1.264009099457 \quad (40)$$

either has a terminating classical collision before the certified terminal section, or its collision-regularized continuation reaches the strict restricted and finite-mass-form escape cones. Hence no collision-free member of (40) is periodic. This is an explicit continuum adjacent to the fourth collision boundary, not merely a certificate for one boundary member.

Proof. Let $[\kappa_4]$ and $[\tau_4]$ be the two MPFR enclosures in the fourth-boundary theorem and set $\bar{\tau} = 37184019/10^8 > \sup[\tau_4]$. Propagate the rational hull of the certified collision state for $\bar{\tau} - [\tau_4]$, obtaining a reference state X_* at a fixed fourth-LC clock. For each parameter tile K , a persistent C^1 propagation on the hull $H = \text{hull}(K, [\kappa_4])$ gives the rigorous mean-value enclosure

$$X(K, \bar{\tau}) \subset X_* + \partial_\kappa X(H, \bar{\tau})(K - [\kappa_4]). \quad (41)$$

Thus no componentwise projection of an implicitly timed fifth-section image is used.

For the negative-primary LC variables $q = U^2, V, h, t$, put

$$R = 9^{1/3}t^{2/3}, \quad C = U/\sqrt{R}, \quad S = \sqrt{1 - C^2}.$$

The exact entry to the simultaneous two-centre chart is regular at $U = 0$:

$$\begin{aligned} p &= -\frac{2}{3}\bar{S}V + \frac{2}{9t}\bar{S}|U|^2U, \\ E &= \frac{Rh}{9} - \frac{4t \operatorname{Re}(U\bar{V})}{3R^2} + \frac{2|U|^4}{9R^2} - \frac{1}{9} \left(\left| \frac{U^2}{R} - \frac{1}{2} \right|^2 + \frac{1}{|S|^2} \right). \end{aligned} \quad (42)$$

At collision these reduce to $p = -2V/3$ and $E = Rh/9 - 5/36$. Direct symbolic substitution using $R^{3/2} = 3t$ verifies (42).

From (41)–(42), a directed Poincare map reaches the first section $\alpha = 2\pi$ with $\beta < 0$. Whole-step enclosures prove $p_r < 0$, so $\alpha_\sigma = -p_r > 0$. After reboxing this six-dimensional image, a second directed map reaches $\beta = -1$ with $p_r < 0 < p_i$ throughout and $\alpha < 3\pi$. Hence no intervening focus is missed. The exact bridge map keeps both light–primary squared distances above 10^{-3} while the signed heavy coordinate crosses zero, and its regularized continuation satisfies both terminal escape inequalities.

Finally use the tiles

$$\kappa(p, r) = \frac{1264009098895 + p}{10^{12}} + [-r, r]10^{-12}$$

with $p = 120, 122, \dots, 198$, $r = 1$, and $p = 200, 204, \dots, 560$, $r = 2$. The 40 and 91 tiles overlap, and their union is exactly (40). Every tile passes the directed-section, whole-step, bridge, and terminal strict inequalities. An earlier zero is a classical collision; conditional on no such zero, uniqueness identifies the regularized and classical arcs up to the isolated heavy collision. This proves the stated dichotomy without any sampling inference.

The aggregate certificate gives, conservatively, first-focus $\beta < -0.0132$, selected squared distance $> 4.4 \times 10^{-5}$, remaining 3π -focus gap > 0.285 , bridge primary-distance square > 0.019 , restricted escape margin > 0.804 , and finite-mass-form margin > 0.935 .

The current interval-Newton projection for the third collision root overlaps (40) but extends beyond its upper endpoint. Consequently this theorem makes no claim that (40) crosses that exact root. $\square$

Theorem 29 (Positive-mass fourth-boundary cusp). *There are a constrained incoming-section neighborhood of the fourth limiting root and $B_0 > 0$ such that every compatible member with $0 < B < B_0$ either suffers a finite classical collision or its light body escapes to infinity. Consequently every collision-free member of this local cusp is nonperiodic.*

Proof. The fourth root selects the primary of mass A . In the finite-mass Jacobi variables put

$$q = Z + \frac{R}{1 + A}, \quad Q = R - \frac{B}{A + B}q.$$

Exact subtraction of the Jacobi equations gives

$$q_{\theta\theta} = -(A + B)\Phi(q) - \Phi(R) - \Phi(q - R)$$

and

$$Q_{\theta\theta} = -\frac{1 + A + B}{A + B} \left[A\Phi \left(Q + \frac{B}{A + B}q \right) + B\Phi \left(Q - \frac{A}{A + B}q \right) \right].$$

Thus $q = u^2$, $dt = |u|^2 d\sigma$ gives a real-analytic collision block at $u = 0$, with preserved constraint

$$2|v|^2 - (A + B) - h|u|^2 = 0.$$

The other-primary separation is strict, so the fourth transverse collision persists on a smooth local sheet.

On the limiting collision-ejection branch, a collision in elliptic two-centre coordinates can occur only at an integral- π focus with $\beta = 0$. The validated replay proves $p_r < 0$, and hence $\alpha_\sigma = -p_r > 0$, on the whole-step enclosures through the next three foci. The strict distances in (39), together with the seventh-focus and bridge clearances, therefore make the segment from a fixed fourth-LC ejection section to a fixed heavy-pair LC entry compact and collision-free. Ordinary continuous dependence transfers that segment. The collision-adapted heavy-pair chart transfers the prescribed binary crossing and the scaled terminal state.

Finally apply the branch-independent hierarchical Jacobi lemma with $\epsilon = 1/100$ and $c = 3/2$. Its limiting terminal margin is greater than 10.3953509156741. The scaled inner energy tends to zero, the future tidal integral is uniformly bounded, and its coefficient is the light mass B . Thus the strict comparison persists for small B . A true collision is classical termination; conditional on no collision, the light body escapes. $\square$

The first root's transversality has an exact positive-mass consequence. In the full late Jacobi equations put $A = \sqrt{1 - B^2}$, $M = 1 + A$, and select the same primary by

$$q = Z - \frac{A}{M}R.$$

Direct subtraction gives

$$q_{\theta\theta} = -(1 + B)\Phi(q) + A\{\Phi(R) - \Phi(q + R)\}.$$

The R equation still contains $B\Phi(q)$, so the common analytic collision chart uses the colliding-pair center coordinate

$$Q = R + \frac{B}{1 + B}q.$$

Then

$$Q_{\theta\theta} = -\frac{M + B}{1 + B} \left[\Phi \left(Q - \frac{B}{1 + B}q \right) + B\Phi \left(Q + \frac{1}{1 + B}q \right) \right].$$

This contains no singular pair force. After $q = u^2$ and $dt = |u|^2 d\sigma$, the equations for (Q, Q_t) acquire an exterior $|u|^2$ and are analytic at $u = 0$ when $Q \neq 0$. Hence the finite-mass LC constraint is

$$2|v|^2 - (1 + B) - h|u|^2 = 0,$$

and $|v|^2 = (1 + B)/2$ at collision. At $B = 0$, $Q = R$; the validated inequalities $R > 2$ and other-primary distance greater than one therefore persist on a sufficiently small common chart.

Corollary 30 (Persistence of the collision boundary). *Fix a regular constrained incoming section before the selected LC block and extend the limiting coordinate κ to a local section coordinate $\hat{\kappa}$. The certified limiting collision continues uniquely under all sufficiently small positive-mass and compatible section-state perturbations δ . The continued root $(\hat{\kappa}_c(B, \delta), \sigma_c(B, \delta))$ is transverse. At fixed (B, δ) its local closest-passage normal form is*

$$r_{\min, \text{local}} = \frac{2D_{B, \delta}^2}{1 + B} (\hat{\kappa} - \hat{\kappa}_c(B, \delta))^2 + O(|\hat{\kappa} - \hat{\kappa}_c(B, \delta)|^3),$$

where $D_{B, \delta} = \det(\partial_{\hat{\kappa}} u, v)$ remains negative.

Proof. The constraint, gauge, and section functions have constant full rank at the certified collision-free base state, so the fixed constrained section is smooth also at $B = 0$; equivalently use the analytic signed- B extension and then restrict to $B \geq 0$. On this section choose a C^1 chart $X_{\text{in}}(B, \hat{\kappa}, \delta)$ which restricts at $(B, \delta) = 0$ to the validated limiting curve. Ordinary pre-LC flow, the strict square-root lift, and the collision-adapted (u, Q) flow make the square shooting map C^1 in $(B, \hat{\kappa}, \delta, \sigma)$. The parameter-dependent implicit-function theorem and the strict determinant bound give existence, local uniqueness, and transversality. Orthogonal projection onto $v^\perp$, followed by $q = u^2$ and $|v|^2 = (1 + B)/2$, gives the displayed coefficient. $\square$

Theorem 31 (Differentiated joint inclination). *Let K be a compact projective-amplitude interval inside the validated pre-collision chart and set $p = \mu/\tau$. In the normalized family $H = \kappa|T|^p$, with stable entrance coordinate S fixed, translate the logarithmic saddle dwell by $L_T = -\tau^{-1} \log |T|$ and scale size by $\rho_T = |T|^{1/\tau}$. If*

$$B|\log |T|| + B + |T|^{2-p} + e^{-cL_T} + \|\hat{\Xi}\| \longrightarrow 0,$$

where $\widehat{\Xi}$ contains the Newtonian-scaled energy, clock, and remaining triangular deviations, then the state and its κ -tangent converge on every common collision-free compact interval. Through the selected limiting collision the same assertion holds in the scaled (u, v, Q, P, h) Levi-Civita chart.

Consequently the certified collision root extends uniquely as a C^1 graph

$$\kappa = \kappa_c(B, T, S, \Xi), \quad \kappa_c(B, T, S, \Xi) \longrightarrow \kappa_c.$$

Its determinant retains the certified negative sign. At fixed parameters,

$$r_{\min, \text{local}} = \rho_T \frac{2\widehat{D}^2}{1+B} (\kappa - \kappa_c)^2 + o(\rho_T |\kappa - \kappa_c|^2).$$

Proof. Choose a fixed weight $\mu/2 < \omega < \tau$. For small B it also satisfies $\mu_B/2 < \omega < \tau_B$, and the quadratic Lyapunov–Perron Green denominator is bounded below by $2\omega - \mu_B > 0$. Construct only the nonlinear unstable-manifold shadow on the infinite weighted tail. Translation changes its two homogeneous coefficients by $e^{(\mu_B - \mu)L_T}$ and $e^{(\tau_B - \tau)L_T}$; these tend to one because $BL_T \rightarrow 0$. Differentiating the uniformly contracting fixed-point equation gives convergence of the κ -tangent.

A bounded finite-entrance stable datum does not belong to that half-line weighted ball. Instead, C^1 stable-foliation holonomy compares the actual orbit and tangent with the shadow at one fixed incoming-tail section, with error $O(e^{-c(L_T - A)} + \rho_T |E|) + o(1)$. Compact variational flow and the zero-angular-momentum fixed-axis reconstruction propagate this estimate. The scaled collision-adapted Q –LC vector field is analytic at $u = 0$ and propagates the variational equation through the regularized zero. The parameter-dependent implicit-function theorem and the certified determinant give the graph. The C^1 expansion of u transverse to the root, together with $q = u^2$ and $|v|^2 = (1+B)/2$, gives the last formula. $\square$

Theorem 32 (Real collision sampling). *There are infinitely many distinct real parameters on the tied normalized Pythagorean curve, accumulating at the skinny endpoint, whose classical solution has a finite-time binary or triple collision.*

Proof. Let χ_* be the asymptotic intercept of the certified transverse intersection between the restricted incoming parabolic curve and the collision-stable curve. In the signed longitudinal amplitude H_0 , choose two compact intercept arcs on opposite sides of χ_* with fixed strict opposite signs. Since

$$\Phi'_{\text{ref}}(B) = -\frac{3\pi}{2} B^{-5/2} (1 + o(1)),$$

every sufficiently late phase winding supplies a disjoint real B -bracket whose reference intercept crosses these arcs. If any member collides before the selected late chart, that bracket already supplies the theorem. If not, uniform incoming-tail convergence and compact regularized flow transfer the two H -signs to the positive-mass saddle entrance.

Put $p = \mu/\tau$. Continuity gives, inside the bracket, a last zero of H followed by a first point at which $H = B^{p/2}$. On the intervening segment $0 \leq H \leq B^{p/2} \rightarrow 0$. Local uniqueness of the transverse restricted intersection therefore forces every sequence in this segment to approach the certified endpoint. Uniformly there,

$$\frac{T_B}{B} \longrightarrow \Theta_* \neq 0, \quad \|\widehat{\Xi}_B\| \longrightarrow 0,$$

so the normalized collision graph is defined. The continuous function

$$G(B) = H_B - |T_B|^p \kappa_c(B, T_B, S_B, \Xi_B)$$

is negative at $H = 0$ because $\kappa_c > 0$, and positive at $H = B^{p/2}$ because $|T_B|^p = O(B^p) = o(B^{p/2})$. Its intermediate-value zero is an exact selected light–heavy collision of the tied positive-mass trajectory. The phase brackets are pairwise disjoint and tend to zero. $\square$

Theorem 33 (Explicit local collision-or-escape cusp). *Every regularized limiting member in*

$$1.2679251755 \leq \kappa \leq 1.2679451755$$

reaches the strict terminal escape cone described below. There is a neighborhood of this closed interval such that every sufficiently small compatible positive-light-mass member either has a finite classical collision before the terminal section or its light body escapes to infinity. Hence every collision-free member in this local cusp has no second brake.

Proof. Ten adjacent pinned interval-boxed propagations cover the full displayed κ -interval, cross any selected collision in its Levi–Civita chart, and reconstruct the restricted outer state at a fixed ejection section. With $\lambda = \sqrt[3]{t}$, a second analytic LC bridge crosses the prescribed heavy-binary collision. Whole-step enclosures keep both light–primary squared distances above 0.89. On the outgoing section they prove the strict restricted escape inequality

$$\dot{\rho}_0 - \frac{2}{2w_0} - 2 - a'(s_0) > 0.54.$$

For the positive-mass transfer, let x be the heavy-pair Jacobi vector, y the light position relative to its center of mass, and put

$$r = |x|, \quad \rho = |y|, \quad w = \rho - r, \quad e = \frac{1}{2}|\dot{x}|^2 - \frac{M}{r}, \quad M = m_a + m_b, \quad \mathcal{M} = M + m_c.$$

The exact Jacobi equations give

$$|\ddot{y}| \leq \frac{\mathcal{M}}{w^2}, \quad |\dot{e}| \leq \frac{2m_c r |\dot{x}|}{w^3}.$$

Fix $\epsilon, c > 0$, set

$$v_b = \sqrt{\frac{2M}{r_0} + 2\epsilon}, \quad K = \int_0^\infty \frac{\sqrt{2M(r_0 + v_b t) + 2\epsilon(r_0 + v_b t)^2}}{(w_0 + ct)^3} dt.$$

If

$$\dot{\rho}_0 - \frac{\mathcal{M}}{cw_0} > v_b + c, \quad e_0 + 2m_c K < \epsilon,$$

then a simultaneous bootstrap gives

$$r(t) \leq r_0 + v_b t, \quad \dot{\rho}(t) > v_b + c, \quad w(t) \geq w_0 + ct, \quad e(t) < \epsilon.$$

Thus only an inner collision can interrupt outer escape. In the certified massless terminal state $M = \mathcal{M} = 2$, $m_c = 0$, and $e_0 = 0$. For $\epsilon = 1/100$, $c = 3/2$, interval propagation proves the stronger full-pair margin

$$\dot{\rho}_0 - \frac{2}{(3/2)(\rho_0 - r_0)} - \sqrt{\frac{4}{r_0} + \frac{2}{100}} - \frac{3}{2} > 0.65.$$

Apply the comparison in the Newtonian-scaled finite-mass terminal coordinates, where $\hat{e} = \rho T e$. Scaled LC convergence gives $\hat{e}_0 \rightarrow 0$, while the correspondingly scaled K is uniformly bounded and $m_c K = O(B)$. The strict inequalities persist. An exact positive-mass collision terminates the classical branch; otherwise the member reaches the terminal section and escapes. $\square$

Corollary 34 (Open-window refinement). *Every sufficiently late skinny phase-cycle bracket either contains a classical collision before the common selected chart, or contains a nonempty open tied parameter interval every member of which collides or escapes.*

Proof. On a bracket without an earlier collision, use the segment in the real collision-sampling proof from $H = 0$ to $H = B^{p/2}$. The flow makes H, T continuous, while the uniform estimate $T/B \rightarrow \Theta_* \neq 0$ keeps the denominator nonzero. Thus the continuous normalized coordinate satisfies

$$\frac{H}{|T|^p} = 0 \quad \text{at the first endpoint,} \quad \frac{H}{|T|^p} \asymp B^{-p/2} \longrightarrow +\infty \quad \text{at the second.}$$

It therefore traverses the interior of $[1.2679251755, 1.2679451755]$. The preimage of that interior is a nonempty open set and hence contains an open interval. Restricted to this preimage, compactness of the projective-amplitude interval, uniform endpoint localization, and differentiated regularized convergence supply the remaining finite-mass cusp hypotheses. The preceding theorem classifies every member of this open set as collision or escape. $\square$

This boundary emanates from the massless planar joint-limit collision and persists in the surrounding finite-mass section family. It is not a tied finite-mass Pythagorean orbit or a brake, and no positive-mass classical collision is continued physically. The preceding theorem classifies a neighborhood of this first root. What remains is exclusion or certification of the remaining collision zeros and outgoing control on the compact components. In particular, the exact tied trajectory selects only one point $(\hat{\kappa}_B, \delta_B)$ on the fixed section; the collision-boundary persistence corollary does not prove that it solves $\hat{\kappa}_B = \hat{\kappa}_c(B, \delta_B)$. The theorem replaces this section-dependent condition by the intrinsic scalar gap

$$\Delta_{\text{intr}}(B) = \frac{H_B}{|T_B|^{\mu/\tau}} - \kappa_c(B, T_B, S_B, \Xi_B).$$

No componentwise sign classification of this gap is proved here. The real collision-sampling theorem gives only a cyclewise dichotomy between an earlier collision and a zero of this selected graph. Its isolated roots are not proved rational.

Parameterize the restricted incoming parabolic curve by its asymptotic phase intercept χ , normalized by $\chi = \lim_{Y \rightarrow \infty} (\phi_Y + (4/3)Y^{3/2}) \bmod 2\pi$. This coordinate includes the finite restricted quadrupole correction between infinity and a fixed section; a raw monopole section phase does not. The incoming-tail theorem inserts the true full regularized state, producing infinitely many open terminal nonperiodicity intervals accumulating at $B = 0$ result. Strict restricted escape on a compact intercept arc transfers to a finite outgoing scaled section; there $e = -C/B + o(B^{-1})$ and $\dot{\rho} = sB^{-1/2} + o(B^{-1/2})$, so the terminal escape lemma applies uniformly, yielding escape or an earlier classical inner collision and excluding a later brake in either case.

Theorem 35 (Infinite subfamily). *There are infinitely many open real Euclid-parameter intervals accumulating at $u = 0$ on which the Pythagorean–Burrau solution is nonperiodic. In particular, infinitely many distinct primitive Pythagorean triples satisfy the nonperiodicity conjecture.*

Proof. Shrink a restricted escape-intercept arc by a fixed positive margin. The incoming-tail theorem and robust terminal transfer cover every sufficiently small preimage interval. A trajectory that collides before section reach is already nonperiodic; every other trajectory satisfies the terminal escape-or-collision certificate. The exact derivative $\Phi'_{\text{ref}} \sim -(3\pi/2)B^{-5/2}$ gives infinitely many disjoint preimage intervals. Since $B(u) = 2u/(1+u^2)$ is a homeomorphism near zero and rational u is dense, each contains a rational Euclid parameter. Reduction by the simultaneous mass–length scaling gives distinct primitive triples. $\square$

The existential rational consequence can be sharpened to one explicit primitive family.

Theorem 36 (Positive density in an explicit family). *Let*

$$(a_n, b_n, c_n) = (4n^2 - 1, 4n, 4n^2 + 1), \quad n \geq 1.$$

There is a positive-lower-natural-density set of indices n for which the corresponding Pythagorean–Burrau solution is nonperiodic.

Proof. Here $u_n = 1/(2n)$ and $B_n = 4n/(4n^2 + 1)$. Direct substitution in the exact reference clock gives

$$F(x) := \frac{\Phi_{\text{ref}}(B_x)}{2\pi} = \frac{(2x - 1)^{3/2}(2x + 1)(4x^2 + 1)^{3/4}}{32x^{5/2}},$$

and therefore

$$F''(x) = \frac{3}{8}x^{-1/2} + \frac{1}{32}x^{-3/2} + O(x^{-5/2}).$$

On each dyadic block, van der Corput’s second-derivative estimate makes every nontrivial Weyl sum $O_h(N^{3/4})$. Weyl’s criterion proves that $F(n)$ is equidistributed modulo one. Consequently every fixed compact escape-intercept arc J selects a set of indices of density $|J|/(2\pi)$. The incoming-tail and terminal-transfer theorems prove every sufficiently large selected member nonperiodic. Primitivity follows from $\gcd(4n, 4n^2 - 1) = 1$. $\square$

9 Cardinality and the remaining obstruction

The rational parameters are countable and dense. Neither countability, measure zero, analyticity, nor the nominal three-equations/two-unknowns count excludes rational zeros. A proof still requires either no real zero, an exact arithmetic characterization of all real zeros, or a finite exact reduction.

The active route factors a possible brake, away from Jacobi degeneracies, into two radial-rate equations followed by an angular-slip equation, and seeks to classify the isolated double-radial events along the first-syzygy continuation. Ordinary scans show both signs of angular slip on different double-radial events, ruling out a universal fixed-sign shortcut. The skinny route now controls the full incoming plunge and proves open escape windows. On the finite-turn side, fixed-resonance first-order matching and a uniform outgoing angular-momentum theorem exclude a brake at the first outer turn throughout the infinite accumulating resonance layer. The returning-tail estimate propagates the sign back to a fixed large section; the next problem is to classify the resulting compact collision-regularized central-encounter map.

10 Necessary integral identities

For a hypothetical second brake at τ , the Lagrange–Jacobi identity is

$$\ddot{I} = 4K - 2U = 2U - 4U_0.$$

Both endpoint dilational momenta vanish, so

$$\int_0^\tau U dt = 2U_0\tau, \quad \int_0^\tau K dt = U_0\tau.$$

Integrating each labelled acceleration also gives zero. If $d_{ij} = q_j - q_i$, multiplication by the fixed initial positions yields the additional scalar condition

$$\int_0^\tau \sum_{i < j} m_i m_j \frac{d_{ij}(0) \cdot d_{ij}(t)}{r_{ij}(t)^3} dt = 0.$$

Its integrand is initially positive, but no global projection inequality has been proved through close encounters.

At energy $-U_0$, the Jacobi–Maupertuis metric is $2(U - U_0)ds_m^2$. A second-brake segment is a degenerate-boundary geodesic chord. A Morse-index or convexity obstruction remains open.

11 Catalog comparison

Li and Liao list a nearby periodic brake orbit on the mass slice $(0.6, 0.8, 1)$. The exact tied point is $(x, y) = (0.14, 0.48)$, while their closest listed entry is $(0.1446319096, 0.4773197126)$. Its side lengths and right-triangle defect do not satisfy the exact equations. Equal-mass modern catalogs are also disjoint from the tied right-triangle curve. No inspected catalog supplies a validated exact counterexample. Local shooting continuation of the F_{30} family varies its masses and reduces the tied-shape gap to about 0.00369, but the two shape-matching coordinates cross zero at distinct mass parameters. This is ordinary numerical evidence, not validated nonintersection.

12 Open proof obligations

The rational and strong real conjectures remain unresolved. The missing theorem must control arbitrarily close collision-free encounters, classify or exclude every double-radial event including Jacobi degeneracies, and be uniform at parameter endpoints. If real intersections remain, an exact arithmetic bridge is still required. In the planar joint-limit family, two transverse collision boundaries are now certified, one for each primary, while the first has a classified local finite-mass collision-or-escape cusp. Global use requires exclusion or certification of the remaining collision zeros and outgoing control on every remaining collision-free component. Differentiated transport through the finite-mass saddle dwell is complete; on the exact tied curve the remaining local condition is the scalar sign/zero problem for $\Delta_{\text{intr}}(B)$. Phase sampling forces infinitely many real classical collisions but does not classify each collision-free component or yield a rational root.

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